

Tail-Index Sure Independence Screening for Ultra-High-Dimensional Heavy-Tailed Regression

Anonymous manuscript

Abstract

We propose a marginal screen for the covariates that determine a positive conditional tail index, designed for problems where the dimension p is large relative to the number of observations informative about the tail. The screen rests on one geometric fact: after projection onto a single coordinate, the conditional distribution is a mixture whose tail is governed by its heaviest component, so the projected tail index is the upper envelope of the tail-index surface along the corresponding fibre. Inactive coordinates produce flat envelopes at the global maximum γ^* ; active coordinates are detectable exactly when their fibres miss the global argmax set on a set of positive measure. Ranking coordinates by the average projected index therefore separates detectable active variables from noise. We estimate the envelope with empirical-rank local Hill profiles, and prove that the maximal score error over p coordinates is $O_{\mathbb{P}}(\sqrt{\log(pn)/(n\alpha h)})$ plus a bias term that our projected-quantile conditions require to be negligible relative to the population gap, where α is the local tail fraction and h the rank bandwidth; $n\alpha h$ is the effective local extreme sample size. Sure screening and exact recovery follow when $\log(pn) = o(n\alpha h\Delta_{\min}^2)$. The algorithm costs $O(pn \log n)$. A rank-aggregated version of the screen, taking the minimum rank over a block of nine neighbouring tunings, stabilizes the ranking against the choice of the smoothing parameters. Simulations across $p \in \{500, 1000, 2000\}$, including a model whose scale is driven by a correlated cluster of twenty inactive covariates, show when screening on the tail index differs from screening on a fixed high quantile: the cluster fills the leading positions of moderate-level quantile screens and displaces the active set by the size of the block, while the aggregated tail screen keeps its leading positions on the active set. An application to violent-crime rates in 1,993 U.S. communities, with $p = 99$ continuous predictors against an effective tail sample of about 200, illustrates the method in its intended regime, and a second application to the maximal token surprisal of a language model over 20,000 web documents, where two scale nuisances have known status — document length by an asymptotic argument, a grafted bounded factor exactly by Breiman’s lemma — observes the same target separation in the rankings, while a permutation-calibrated familywise test declares no discovery in either direction. The uniform projected-quantile conditions are stated directly; deriving them from a fully primitive growing-dimensional model remains open.

1 Introduction

Let (X, Y) be a random pair with $Y > 0$ and $X \in \mathbb{R}^p$, and suppose that, conditionally on $X = x$,

$$\frac{\mathbb{P}(Y > ty \mid X = x)}{\mathbb{P}(Y > y \mid X = x)} \longrightarrow t^{-1/\gamma(x)}, \quad y \rightarrow \infty, \quad t > 0, \quad (1)$$

for a positive function γ , the conditional tail index. Estimating γ is the first step toward extreme conditional quantiles and tail probabilities [6, 12], and it suffers from a particularly severe curse of dimensionality: local estimation must find observations near x and then keep only an upper fraction of their responses. A one-dimensional window of width h combined with a tail fraction α leaves

an effective sample of order $n\alpha h$, not n [5, 10]. When p is large, even moderate sample sizes are effectively small.

This paper asks the screening question: among p candidate covariates, possibly with p growing with n , which ones does γ actually depend on? Sure independence screening [7] answers such questions by ranking coordinates through inexpensive marginal statistics and keeping a moderate number for subsequent joint analysis. Model-free variants exist for general dependence [13] and for fixed conditional quantiles [11], but the limiting exponent in (1) is a different target: a covariate can move the 0.95 quantile of Y without affecting γ , and conversely. Our simulations make this distinction concrete.

The screen we propose rests on a single geometric mechanism. Transform each continuous covariate to $U_j = F_j(X_j)$ and let $\xi_j(u)$ denote the tail index of Y given $U_j = u$ alone. Conditioning on one coordinate leaves the others free to range over the fibre $\{u\} \times [0, 1]^{p-1}$, so the projected distribution is a mixture, and a mixture of heavy tails is governed by its heaviest component. Under a full-support condition,

$$\xi_j(u) = \max_{v \in [0, 1]^{p-1}} \gamma(u, v) : \quad (2)$$

the projected index is the upper envelope of the tail-index surface along the fibre. If coordinate j is inactive, every fibre reaches the global maximum $\gamma^* = \max_x \gamma(x)$ and the envelope is flat at γ^* . If coordinate j is active and some of its fibres miss the global argmax set, the envelope dips below γ^* there. Averaging the envelope over a trimmed interval gives a score Ψ_j whose gap $\Delta_j = \gamma^* - \Psi_j$ is zero for every inactive coordinate and positive for every detectable active one. Ranking coordinates by increasing Ψ_j is the screen.

The same geometry states the method's limitation. An active coordinate is invisible when every one of its fibres still meets the global argmax set; this happens, for instance, for $\gamma(u_1, u_2) = \gamma_0 - (u_1 - u_2)^2$, whose argmax is the diagonal. Detectability is a property of the argmax projections, and we characterize it exactly rather than assuming a generic signal-strength condition.

Estimation follows the envelope. For each coordinate we slide a window over the empirical marginal ranks, apply a Hill statistic to the upper α -fraction of the responses in the window, and average the resulting profile. Empirical ranks give every coordinate the same deterministic window geometry and remove the marginal scales. The price is a rank-displacement term, which we control simultaneously over p coordinates by a Dvoretzky–Kiefer–Wolfowitz bound; it is negligible when $\log(pn) = o(nh^2)$. The main result bounds the maximal score error by

$$O_{\mathbb{P}} \left(\sqrt{\frac{\log(pn)}{n\alpha h}} \right) + o(\Delta_{\min}),$$

where $\Delta_{\min}$ is the weakest population gap, and yields sure screening when $\log(pn) = o(n\alpha h \Delta_{\min}^2)$: ultra-high dimension is affordable exactly when the local extreme count dominates the logarithm of the dimension. A blockwise implementation costs $O(pn \log n)$ time.

Our construction adapts the projected-tail framework of Gardes and Podgorny [9], who estimate a low-dimensional central tail-index subspace by minimizing an average of projected local Hill estimates over projection matrices, in fixed dimension. We keep their projection semantics and local Hill estimator, and change what is needed for screening: the projections are the p coordinates themselves, p may diverge, windows are built from empirical ranks with deterministic local counts, and uniformity over a compact matrix class is replaced by union bounds over coordinates and window states. Their density assumptions on the covariate are not needed here; the empirical-rank construction replaces them. Among screening proposals, Yoshida and Umezu [19] screen on covariate-dependent extreme-value indices through kernel conditional Pickands contrasts in a

large-bandwidth regime, and parametric tail-index regression [18], extended to high dimensions with regularization by Sasaki et al. [17], exploits a parametric link when one is available. Tail dimension reduction methods [1, 2, 4, 8] target related but distinct conditional-tail objects.

The theoretical scope deserves a clear statement. The primitive model (1) identifies the projected index (2), but does not by itself provide the uniform projected-quantile regularity needed for growing-dimensional concentration: projection can create additional logarithmic slow variation even from exact Pareto conditionals. We therefore state the required conditions directly on the projected quantiles and leave their derivation from primitive assumptions open.

Section 2 develops the population theory. Section 3 defines the estimator and states the screening results. Section 4 presents the simulation models, the tuning study together with a rank-aggregated version of the screen built on a block of nine neighbouring tunings, and a comparison with competing screens at $p \in \{500, 1000, 2000\}$ that covers, in the same tables, the sensitivity to the dimension and to the quantile level of fixed-quantile screening. Section 5 applies the aggregated screen to violent-crime rates in U.S. communities. Section 6 builds a second application, on the maximal token surprisal of a language model, in which two scale-nuisance clusters have known status and the screening is calibrated by permutation. Section 7 discusses limitations. Proofs are in Appendix B, supported by the auxiliary results of Appendix A.

2 Population theory

2.1 Model and assumptions

All quantities below may depend on n : the dimension $p = p_n$, the tail-index function γ , the active set A , its size $s = |A|$, and later the bandwidth h , the tail fraction α and the retained size d . The index n is suppressed whenever no ambiguity arises.

Notation. A positive measurable function f is slowly varying at infinity if $f(ty)/f(y) \rightarrow 1$ as $y \rightarrow \infty$ for every $t > 0$, and a survival function $\bar{F}$ is regularly varying with index $-1/\xi < 0$ if $\bar{F}(ty)/\bar{F}(y) \rightarrow t^{-1/\xi}$ for every $t > 0$. Generalized inverses are $F^{-1}(t) = \inf\{y : F(y) \geq t\}$. For $u \in [0, 1]$ and $v \in [0, 1]^{p-1}$, $\iota_j(u, v) \in [0, 1]^p$ denotes the vector with u in coordinate j and v in the remaining coordinates, in order; we abbreviate $\gamma(u, v) = \gamma\{\iota_j(u, v)\}$ when j is clear from context. Norms on $[0, 1]^A$ are Euclidean. All asymptotic statements concern a triangular array indexed by n , and $Z_n = O_{\mathbb{P}}(a_n)$ means $\lim_{C \rightarrow \infty} \limsup_n \mathbb{P}(|Z_n| > Ca_n) = 0$.

The observed data are i.i.d. pairs (X_i, Y_i) with $Y_i > 0$ and $X_i \in \mathbb{R}^p$ with continuous marginal distribution functions $F_1, \dots, F_p$. The theory is formulated on the latent probability-integral transforms $U_{ij} = F_j(X_{ij})$, so that $U_i \in [0, 1]^p$ has uniform one-dimensional marginals and unrestricted cross-coordinate dependence; the transforms are never needed by the algorithm, because the ranks of $X_{1j}, \dots, X_{nj}$ coincide with the ranks of $U_{1j}, \dots, U_{nj}$ almost surely. With slight abuse of notation we write γ for the tail-index function on the cube, $\gamma(u) = \gamma_X\{F_1^{-1}(u_1), \dots, F_p^{-1}(u_p)\}$, where γ_X is the function appearing in (1).

(C1) *Conditional regular variation.* For every u , the conditional survival function factorizes as

$$\bar{F}(y | u) = \mathbb{P}(Y > y | U = u) = y^{-1/\gamma(u)} L(y, u),$$

where $L(\cdot, u)$ is slowly varying at infinity, $\gamma : [0, 1]^p \rightarrow (0, \infty)$ is continuous, and $0 < \underline{\gamma} \leq \gamma(u) \leq \Gamma < \infty$ uniformly in n and u .

(C2) *Uniform tail regularity.* There is a function c with $0 < c_- \leq c(u) \leq c_+ < \infty$ such that $\sup_u |L(y, u)/c(u) - 1| \rightarrow 0$ as $y \rightarrow \infty$.

(C1) states that each conditional law is heavy-tailed with index $\gamma(u)$, as in (1); (C2) strengthens the slow variation of L to uniform convergence toward a bounded positive limit, the form needed to control mixtures of the conditional laws.

The active set A collects the coordinates on which γ genuinely depends: $j \in A$ if and only if there exist u, v differing only in coordinate j with $\gamma(u) \neq \gamma(v)$. Iterating over the complementary coordinates shows that $\gamma(u) = g(u_A)$ for some function g , and that A is the smallest coordinate set with this property. Membership in A concerns the limiting exponent only, so a coordinate may shift the body of the conditional distribution, or its second-order behaviour, without being active. Conditions (C1)–(C2) parallel conditions (C.1)–(C.2) of Gardes and Podgorny [9], except that we do not assume the law of $\gamma(U)$ to be absolutely continuous; the projection argument below handles atoms, which arise as soon as one coordinate is inactive.

Fix a trimming parameter $0 < \varepsilon < 1/2$ and write $I_\varepsilon = [\varepsilon, 1 - \varepsilon]$. For each coordinate j , fix a pointwise conditional distribution $K_j(u, dv)$ of U_{-j} given $U_j = u$, and define the conditional law of Y given $U_j = u$ to be compatible with it through the disintegration identity

$$\mathbb{P}(Y > y \mid U_j = u) = \int \mathbb{P}\{Y > y \mid U = (u, v)\} K_j(u, dv);$$

without such pointwise versions the statements below hold for almost every u . Define

$$\xi_j(u) = \text{ess sup}_{v \sim K_j(u, \cdot)} \gamma(u, v),$$

the essential supremum of $v \mapsto \gamma(u, v)$ under $K_j(u, \cdot)$; Proposition 2.1 shows that the conditional law of Y given $U_j = u$ is regularly varying with index $-1/\xi_j(u)$, so $\xi_j(u)$ is its tail index. One further condition is used only where stated:

- (S) *Full fibre support.* For every j and $u \in I_\varepsilon$, the support of $K_j(u, \cdot)$ is all of $[0, 1]^{p-1}$. A continuous joint density, positive on the interior cube, is sufficient.

2.2 Projection turns the surface into an envelope

Conditioning on $U_j = u$ mixes the conditional laws of Y over the fibre $\{u\} \times [0, 1]^{p-1}$. Heuristically, the slowest-decaying component of a mixture of heavy tails dictates the tail of the mixture, so the projected index should equal the largest tail index available on the fibre, and a localized region of heavy tails should survive projection intact. The following proposition makes this precise.

For a sub- σ -field $\mathcal{G}$, the conditional essential supremum $\text{ess sup}\{\gamma(U) \mid \mathcal{G}\}$ is the smallest $\mathcal{G}$ -measurable random variable dominating $\gamma(U)$ almost surely.

Proposition 2.1 (Projection). *Assume (C1)–(C2) and let $\mathcal{G} \subseteq \sigma(U)$. Then $\mathbb{P}(Y > y \mid \mathcal{G})$ is almost surely regularly varying with index $-1/\xi$, where $\xi = \text{ess sup}\{\gamma(U) \mid \mathcal{G}\}$. In particular $\xi_j(u)$ is the conditional essential supremum of $\gamma(U)$ given $U_j = u$. Under (S),*

$$\xi_j(u) = \max_{v \in [0, 1]^{p-1}} \gamma(u, v), \quad u \in I_\varepsilon. \quad (3)$$

The essential-supremum identity requires neither full support nor absolute continuity; (S) enters only to replace the essential supremum by the maximum over the whole fibre. If dependence removes part of a fibre, the maximum runs over the conditional support actually available at u , and the population target changes accordingly.

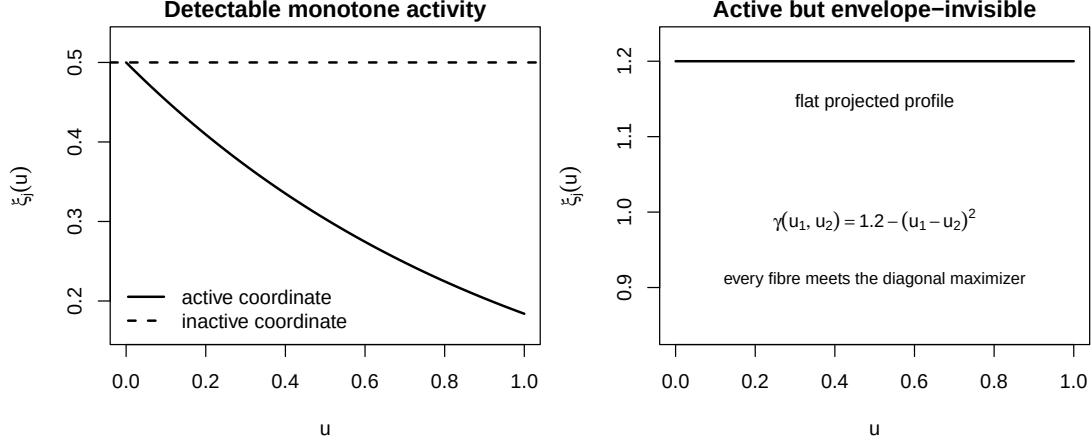


Figure 1: Projected envelopes. Left: a monotone active coordinate depresses the envelope below the inactive baseline γ^* . Right: in the diagonal example every fibre meets the argmax set, so the envelope of an active coordinate stays flat.

2.3 Score, gap, and detectability

Write $\gamma^* = \max_u \gamma(u)$ for the global maximum and

$$\Psi_j = \frac{1}{1-2\varepsilon} \int_{I_\varepsilon} \xi_j(u) du, \quad \Delta_j = \gamma^* - \Psi_j, \quad \Delta_{\min} = \min_{j \in A} \Delta_j. \quad (4)$$

The screen ranks coordinates by increasing Ψ_j and keeps the d smallest. Small scores indicate activity: all inactive coordinates share the flat baseline γ^* , while fixing a detectable active coordinate lowers the attainable envelope. The algorithm uses only the ordering of the Ψ_j and never estimates γ^* .

Let $M = \arg \max_u \gamma(u)$, a nonempty compact set, and let $\pi_j(M)$ be its projection onto coordinate j .

Proposition 2.2 (Detectability). *Assume (C1)–(C2) and (S).*

1. If $j \notin A$, then $\xi_j \equiv \gamma^*$ on I_ε and $\Delta_j = 0$.
2. For every j and $u \in I_\varepsilon$: $\xi_j(u) = \gamma^*$ if and only if $u \in \pi_j(M)$.
3. $\Delta_j > 0$ if and only if $I_\varepsilon \setminus \pi_j(M)$ has positive Lebesgue measure.

If the restriction of γ to the active coordinates has a finite set of maximizers, every active coordinate is detectable.

Claim 3 separates two failure modes that finite-sample experiments tend to blur. When $\pi_j(M)$ covers I_ε , no sample size makes coordinate j identifiable from this score; the diagonal example $\gamma(u_1, u_2) = \gamma_0 - (u_1 - u_2)^2$, with $M = \{(t, t) : 0 \leq t \leq 1\}$, is of this kind, and both of its active coordinates are invisible. When $\pi_j(M)$ misses a set of positive measure, detection is a question of estimation accuracy, quantified by Section 3. Figure 1 contrasts the two situations.

Proposition 2.3 (Quantitative margin). *Assume (C1)–(C2), (S), and that for some $u_A^* \in [0, 1]^A$, $c > 0$, $\nu > 0$,*

$$\gamma(u) \leq \gamma^* - c \|u_A - u_A^*\|^\nu \quad \text{for all } u.$$

Then, for $j \in A$,

$$\gamma^* - \xi_j(u) \geq c|u - u_j^*|^\nu, \quad \Delta_j \geq c\mathbb{E}(|V - u_j^*|^\nu), \quad V \sim \text{Unif}(I_\varepsilon).$$

For $\nu = 2$ the right-hand side is at least $c\{(1 - 2\varepsilon)^2/12 + (1/2 - u_j^*)^2\}$.

Two consequences of (3) help with examples. If $\gamma(u) = G(\sum_{\ell \in A} g_\ell(u_\ell))$ with G strictly increasing, then $\xi_j(u) = G(g_j(u) + \sum_{\ell \in A, \ell \neq j} \max g_\ell)$, and $\Delta_j > 0$ whenever g_j lies below its maximum on a subset of I_ε of positive measure. If $\partial_j \gamma \geq c_j > 0$ throughout the cube, arbitrary interactions are allowed and $\Delta_j \geq c_j/2$.

For the benchmark $\gamma(u) = \frac{1}{2} \exp(-u_1 - u_2 - u_3 - u_4)$ used in Section 4, $\xi_j(u) = e^{-u}/2$ for $j \leq 4$ and $\xi_j \equiv 1/2$ otherwise, so $\Delta_j \approx 0.186$ at $\varepsilon = 0.05$. The gap is comfortable, yet estimating $\xi_j(u)$ near its maximum requires sample points whose remaining three active coordinates are simultaneously small; the simulations keep population detectability and finite-sample accessibility clearly separated.

3 Estimation and screening theory

3.1 Empirical-rank local Hill scores

The estimator follows the envelope construction coordinate by coordinate. Let R_{ij} be the rank of X_{ij} among $X_{1j}, \dots, X_{nj}$ and $\widehat{U}_{ij} = R_{ij}/(n+1)$. For $u \in I_\varepsilon$, form the rank window

$$\mathcal{W}_j(u) = \{i : |\widehat{U}_{ij} - u| \leq h\}, \quad M_j(u) = |\mathcal{W}_j(u)|, \quad k_j(u) = \lfloor \alpha M_j(u) \rfloor,$$

and, writing $Y_{j,u,(1)} \leq \dots \leq Y_{j,u,(M_j(u))}$ for the local order statistics, the local Hill estimate

$$\widehat{\xi}_j(u) = \frac{1}{k_j(u)} \sum_{r=1}^{k_j(u)} \log \frac{Y_{j,u,(M_j(u)-r+1)}}{Y_{j,u,(M_j(u)-k_j(u))}}, \quad (5)$$

defined when $\alpha M_j(u) > 1$. This is the local-fraction Hill statistic of Gardes and Podgorny [9, Definition 5] evaluated on one-coordinate rank windows. The score averages the profile over the deterministic grid $G = \{r/(n+1) : r = 1, \dots, n\} \cap I_\varepsilon$,

$$\widehat{\Psi}_j = \frac{1}{|G_j^{\text{val}}|} \sum_{u \in G_j^{\text{val}}} \widehat{\xi}_j(u), \quad G_j^{\text{val}} = \{u \in G : \alpha M_j(u) > 1\}, \quad (6)$$

with $\widehat{\Psi}_j = +\infty$ when no grid point is usable, so that underpopulation can never favour a coordinate. Because the window counts $M_j(u)$ depend on the ranks only through the grid position, G_j^{val} is in fact the same set for every j ; under the conditions of Theorem 3.1 every grid point is usable and $G_j^{\text{val}} = G$. The screened set $\widehat{A}_d$ collects the d smallest scores, ties broken by coordinate index.

The two localization parameters play different roles: h fixes how many observations resemble the target rank position, roughly $2nh$ of them, and α fixes how many of their largest responses enter the Hill statistic, roughly $2nah$. Rank windows make these counts deterministic up to boundary effects, which is what permits exact finite-sample union bounds over a diverging number of coordinates.

3.2 Assumptions

Ranks displace the unobserved transforms U_{ij} , windows displace the conditioning point, and the Hill statistic sees only the upper α -fraction of a projected distribution. The assumptions bound these

three effects uniformly over the coordinates. Throughout this section every active coordinate is assumed detectable, $\Delta_{\min} > 0$ for every n , so that the conditions below are well posed. Let $Q_j(\cdot, u)$ denote the generalized upper conditional quantile of Y given $U_j = u$, $Q_j(s, u) = F_{Y|j}^{-1}(1 - s^{-1} | u)$ with $F^{-1}(t) = \inf\{y : F(y) \geq t\}$, and let $\tau_n = (pn)^{-2}$.

(E1) *Quantile representation and regularity of the projected index.* For every $j \leq p$ and every $u \in [0, 1]$ the conditional distribution function $F_{Y|j}(\cdot | u)$ is continuous, so that $V_{ij} = 1 - F_{Y|j}(Y_i | U_{ij})$ is conditionally standard uniform and $Y_i = Q_j(1/V_{ij}, U_{ij})$ almost surely. Moreover the projected indices are equicontinuous on I_ε , uniformly over coordinates and over n ,

$$\omega_\xi(r) = \max_{j \leq p} \sup_{u, v \in I_\varepsilon, |u-v| \leq r} |\xi_j(u) - \xi_j(v)| \longrightarrow 0 \quad (r \rightarrow 0),$$

and satisfy the grid-scale regularity condition $\omega_\xi(1/n) = o(\Delta_{\min})$.

(E2) *Uniform tail regularity.* By Proposition 2.1 and inversion (Lemma A.7), $Q_j(\cdot, u)$ is regularly varying with index $\xi_j(u)$, so it admits the representation $Q_j(s, u) = s^{\xi_j(u)} \ell_j(s, u)$ with $\ell_j(\cdot, u)$ slowly varying; we require

$$\max_{j \leq p} \sup_{u \in I_\varepsilon} \sup_{s \geq (3\alpha)^{-1}, t > 1} \frac{|\log\{\ell_j(ts, u)/\ell_j(s, u)\}|}{\log t} = o(\Delta_{\min}).$$

(E3) *Uniform local quantile regularity.* There exists a constant $C_0 > 1$ such that

$$\max_{j \leq p} \sup_{u \in I_\varepsilon} \sup_{v \in [0, 1], |u-v| \leq C_0 h} \sup_{w \in [\tau_n, 1 - \tau_n]} \left| \log \frac{Q_j(w^{-1}, v)}{Q_j(w^{-1}, u)} \right| = o(\Delta_{\min}).$$

The theorems use only this one C_0 ; under (7) any $C_0 > 1$ suffices.

(E1) asks for continuity only; strict monotonicity of the conditional distribution function is not needed, because flat pieces are visited with probability zero. (E2) is the analogue of condition (H.4) in Gardes and Podgorny [9], normalized by $\log t$ and required only beyond the working tail level $(3\alpha)^{-1}$; it accommodates slowly varying factors such as $\ell(s) \asymp (\log s)^{-2}$, for which the left side is of order $1/\log(1/\alpha) \asymp 1/\log n$ when $\alpha = n^{-a}$, so gaps $\Delta_{\min} \gg 1/\log n$ are covered and smaller gaps are not. (E3) adapts their condition (H.5): the truncation $\tau_n = (pn)^{-2}$ makes the probability that any of the pn conditional PIT variables leaves $[\tau_n, 1 - \tau_n]$ vanish, which is the high-dimensional replacement for their fixed-dimension truncation. Their density conditions (H.1)–(H.2) have no counterpart here: empirical-rank windows have deterministic counts, so no lower bound on a covariate density is required.

These conditions are stated on the projected quantiles, not derived from (C1)–(C2): projection can generate genuinely new slow variation. If $W \sim U(0, 1)$ and $\mathbb{P}(Y > y | W) = y^{-1/(1+W)}$ exactly, then $\mathbb{P}(Y > y) \sim 4y^{-1/2}/\log y$, so even exact Pareto conditionals produce a logarithmic factor after projection, and the left side of (E2) is then of exact order $1/\log(1/\alpha)$. Exhibiting primitive model classes whose coordinatewise projections satisfy (E1)–(E3) with useful rates is left open, together with the derivation of the conditions themselves from a fully primitive model.

3.3 Main results

Theorem 3.1 (Score concentration). *Assume (C1)–(C2), (S), (E1)–(E3), $\alpha \leq 1/3$, $h \leq 1 - \varepsilon$, and*

$$\log(pn) = o(nh^2), \quad \log(pn) = o(n\alpha h). \quad (7)$$

Then there are deterministic sequences $a_n = O(\sqrt{\log(pn)/(n\alpha h)})$ and $b_n = o(\Delta_{\min})$ with

$$\mathbb{P}\left(\max_{j \leq p} |\hat{\Psi}_j - \Psi_j| \leq a_n + b_n\right) \rightarrow 1;$$

in shorthand, $\max_{j \leq p} |\hat{\Psi}_j - \Psi_j| = O_{\mathbb{P}}(\sqrt{\log(pn)/(n\alpha h)}) + o(\Delta_{\min})$.

The stochastic term is the exponential-spacings fluctuation of $O(n)$ local Hill statistics per coordinate, each built from at least $c n \alpha h$ spacings, controlled simultaneously over p coordinates; $\sqrt{\log(pn)/(n\alpha h)}$ is the resulting effective-tail-sample-size rate. The $o(\Delta_{\min})$ term collects the tail bias (E2), the spatial bias (E3) evaluated at $C_0 h$ — the first condition in (7) makes the rank displacement smaller than $(C_0 - 1)h$, so windows never leave the range covered by (E3) — and the quadrature error $\omega_{\xi}(1/n) + O(1/n)$ from (E1).

Theorem 3.2 (Sure screening). *Assume the conditions of Theorem 3.1 (including $\Delta_{\min} > 0$, i.e. every active coordinate detectable in the sense of Proposition 2.2) and*

$$\log(pn) = o(n\alpha h \Delta_{\min}^2). \quad (8)$$

Then, for every $d \geq s$,

$$\mathbb{P}(A \subseteq \hat{A}_d) \rightarrow 1,$$

and with probability tending to one every active score is smaller than every inactive score.

Corollary 3.3 (Exact recovery). *Under the conditions of Theorem 3.2, if $d = s$ then $\mathbb{P}(\hat{A}_s = A) \rightarrow 1$.*

The proofs are in Appendix B, supported by the auxiliary results of Appendix A, whose centrepiece is a finite-sample uniform profile bound (Proposition A.14). The roadmap: (i) a Dvoretzky–Kiefer–Wolfowitz event per coordinate controls all empirical-rank displacements simultaneously, contributing the $\log(pn) = o(nh^2)$ requirement; (ii) rank windows have deterministic local counts and pass through at most $4n + 1$ membership states per coordinate; (iii) the pn conditional probability-integral transforms are truncated to $[\tau_n, 1 - \tau_n]$ at probability cost $2pn\tau_n$, which is what fixes $\tau_n = (pn)^{-2}$ in (E3); (iv) conditionally on a covariate column, the Rényi representation turns each local Hill statistic into a mean of i.i.d. exponentials built from about $n\alpha h$ extremes, plus a slowly varying remainder bounded through (E2); (v) a union bound over the $p(4n + 1)$ states yields the $\log(pn)$ factor; (vi) a deterministic quadrature step converts profile accuracy into score accuracy at cost $\omega_{\xi}(1/n) + O(1/n)$; (vii) uniform score accuracy below $\Delta_{\min}/2$ gives deterministic separation of active from inactive scores. Gardes and Podgorny [9, Theorem 3] establishes the corresponding uniform consistency in fixed dimension over a compact class of projection matrices; their covering-number argument and density assumptions are replaced here by the rank construction, and their result does not imply the growing- p bound.

Condition (8) quantifies the cost of dimension: screening p coordinates costs $\log p$ against the effective local extreme count $n\alpha h$, deflated by the squared gap. With $\alpha = n^{-a}$ and $h = n^{-b}/2$ and $b \leq a$, dimensions up to $\log p = o(n^{1-a-b}\Delta_{\min}^2)$ are admissible — a range of p growing as $\exp\{o(n^{1-a-b}\Delta_{\min}^2)\}$, nearly exponential in a power of n for fixed gaps; for $b > a$ the rank-displacement condition $\log(pn) = o(nh^2) = o(n^{1-2b})$ binds first. These ranges treat (E3) as given; a spatial modulus of order $h \log(1/\tau_n)$, the natural benchmark, adds the constraint $\log(pn) = o(n^b)$.

4 Simulation study

Each coordinate is scored in $O(n \log n)$ time: one sort by empirical rank, then a sliding window over the rank grid whose local Hill statistics are maintained by a Fenwick tree indexed by global

response ranks. Covariates are generated and processed in blocks, so the total cost is $O(pn \log n)$ time and $O(n)$ working memory beyond the scores. At $(n, p) = (2000, 1000)$ one complete screen takes under a second on a laptop core. Scores are sorted increasingly, ties broken by coordinate index; a coordinate whose grid is entirely underpopulated receives score $+\infty$.

4.1 Simulation models

Family A: tail-index structure. Models A1, A2 and A3 share the covariate design, the slowly varying factor, and the tail-index active set $A_\gamma = \{1, 2, 3, 4\}$; they differ only in the tail-index function. Covariates are latent Gaussian AR(1),

$$Z_{i1} \sim N(0, 1), \quad Z_{ij} = \rho Z_{i,j-1} + \sqrt{1 - \rho^2} \epsilon_{ij}, \quad j = 2, \dots, p,$$

transformed to $U_{ij} = \Phi(Z_{ij})$; the tail-index functions are evaluated at the population transforms U , while the estimator sees only Z and computes its own empirical ranks. Responses are generated by quantile inversion, $Y = V^{-\gamma(U)} \ell(1/V, \cdot)$ with $V \sim \text{Unif}(0, 1)$, the mechanism of Gardes and Podgorny [9], and all three models carry their slowly varying factor

$$\ell_1(s, u) = [1 + \exp\{v(u) - s\}]^{-1}, \quad v(u) = 2.5(u_5 + u_6 + u_7 + u_8),$$

which satisfies $\ell_1(s, u) \rightarrow 1$ as $s \rightarrow \infty$: its effect vanishes in the limit while perturbing estimation at finite tail levels. The tail-index functions are

$$\begin{aligned} \gamma_{A1}(u) &= 0.5 \exp(-\sum_{j=1}^4 u_j), \\ \gamma_{A2}(u) &= 0.5 \exp(-0.5 \sum_{j=1}^4 u_j), \\ \gamma_{A3}(u) &= 0.5 \exp(-0.80 \sum_{j=1}^4 u_j - 0.20 u_1 u_2 - 0.15 u_3 u_4). \end{aligned}$$

A1 is the monotone reference, with trimmed population gap 0.186 (Section 2). A2 weakens the signal, with gap 0.107. A3 adds interactions among the active coordinates while keeping a reasonably strong signal, with gap 0.158. All three share the maximum $\gamma^* = 0.5$.

Family B: finite-quantile scale contamination. Model B1 keeps the tail-index function of A1, $\gamma_{B1} = \gamma_{A1}$, hence the same active set and gap, and *replaces* ℓ_1 by a scale factor that persists at every finite quantile (the coordinates ℓ_1 would read are overwritten by the proxy block below, so B1 carries no ℓ_1).

Let $F_i \sim N(0, 1)$ be independent of the active coordinates and of the base AR(1) innovations. The response uses

$$\log \ell_2(s, F) = -\frac{1}{2s} + 0.5 \{\Phi(F) - 0.5\},$$

so the persistent scale factor lies in $[e^{-0.25}, e^{0.25}]$, strictly positive and uniformly bounded. The factor F is not observed. Instead, a block of $q_{\text{scale}} = 20$ proxies occupies the coordinates $A_{\text{scale}} = \{5, \dots, 24\}$:

$$Z_{ij} = 0.7 F_i + \sqrt{1 - 0.7^2} \epsilon_{ij}, \quad j \in A_{\text{scale}},$$

with independent standard normal proxy noises, so each proxy is marginally standard normal, loads on F at 0.7, and the pairwise correlation within the block is 0.49. The B1 covariate vector is built from a base AR(1) vector of length p : coordinates 1–4 and 25– p are kept from the base vector, coordinates 5–24 are replaced by the proxies, and the AR(1) chain beyond coordinate 24 continues from the base coordinate 24, not from the overwritten proxy. The observed B1 covariance

is therefore deliberately not AR(1) over all coordinates; the AR(1) background is intact outside the proxy block.

In B1, variables 1–4 determine the limiting tail index; variables 5–24 are informative about the finite conditional scale through their common information about F , and do not change the limiting exponent. A screen built on a fixed high quantile can therefore legitimately rank the proxy cluster highly — those variables move its target — even though the cluster is inactive for tail-index screening. B1 measures the practical consequence of this target difference.

Monte Carlo design. The common design is

$$n = 2000, \quad p \in \{500, 1000, 2000\}, \quad \rho = 0.25, \quad d = 20, \quad \varepsilon = 0.05,$$

with 1,000 replications per cell and per model in the method comparison (binomial standard errors at most 0.016) and 200 common-random-number replications per model in the tuning study (standard errors at most 0.036); Sure- d denotes $\mathbb{P}(R_{\max} \leq d)$ with $R_{\max} = \max_{j \in A_\gamma} \text{rank}(j)$, and all methods within a replication are evaluated on the same dataset.

4.2 Tuning and the aggregated screen

The two smoothing parameters are parameterized as

$$\alpha = n^{-a}, \quad h = n^{-b}/2,$$

following the convention of Gardes and Podgorny [9]. Sure-20 is evaluated for each model over the crossed grid $a \in \{.25, .30, .35, .40, .45, .50\}$, $b \in \{0, .05, .10, .15, .20, .30, .40\}$ at $n = 2000$, $p = 1000$, $\rho = 0.25$, on 200 common-random-number replications per model: the 200 datasets are generated once per model and every cell of the grid is evaluated on the same datasets, so that differences between cells are not inflated by simulation noise. Figure 2 shows the four heatmaps.

The row $a = 0.25$ collapses for the three A-family models (Sure-20 at most .125, .080 and .225): at $\alpha = n^{-0.25} \approx 0.15$ the local Hill statistics reach into quantile regions where ℓ_1 , whose argument $v(u) - s$ is dominated by the U_5 – U_8 combination at moderate s , deforms the projected distributions. B1, whose bounded scale factor produces no such deformation, degrades only mildly there (up to .580). Performance also falls toward large a and large b together, where the local extreme count collapses, and the degenerate bandwidth $b = 0$ is never competitive. In between, the six cells $a \in \{0.30, 0.35\}$, $b \in \{0.10, 0.15, 0.20\}$ contain every model’s best or near-best performance, with four-model means between 0.718 and 0.780; the pair

$$(a^*, b^*) = (0.30, 0.15)$$

maximizes the four-model mean on the grid (0.780), with per-model Sure-20 .935, .555, .955 and .675, and we adopt it as the single setting. The region is one of good rather than uniform performance: A1, A3 and B1 move little across the six cells, while the weak-signal A2 drops from .555 to .360 between (0.30, 0.15) and (0.35, 0.20), a paired difference well outside Monte Carlo error under the common-random-number design. Because the same 200 replications select the pair and describe the grid, the reported per-cell values carry the usual selection optimism at the margin; the comparison of Section 4.3 uses an independent 1,000-replication stream. At $n = 2000$ the selected pair gives $\alpha \approx 0.10$, $h \approx 0.16$, and about $2n\alpha h = n^{1-a^*-b^*} \approx 65$ effective local extremes per interior window. No model is given its own tuning.

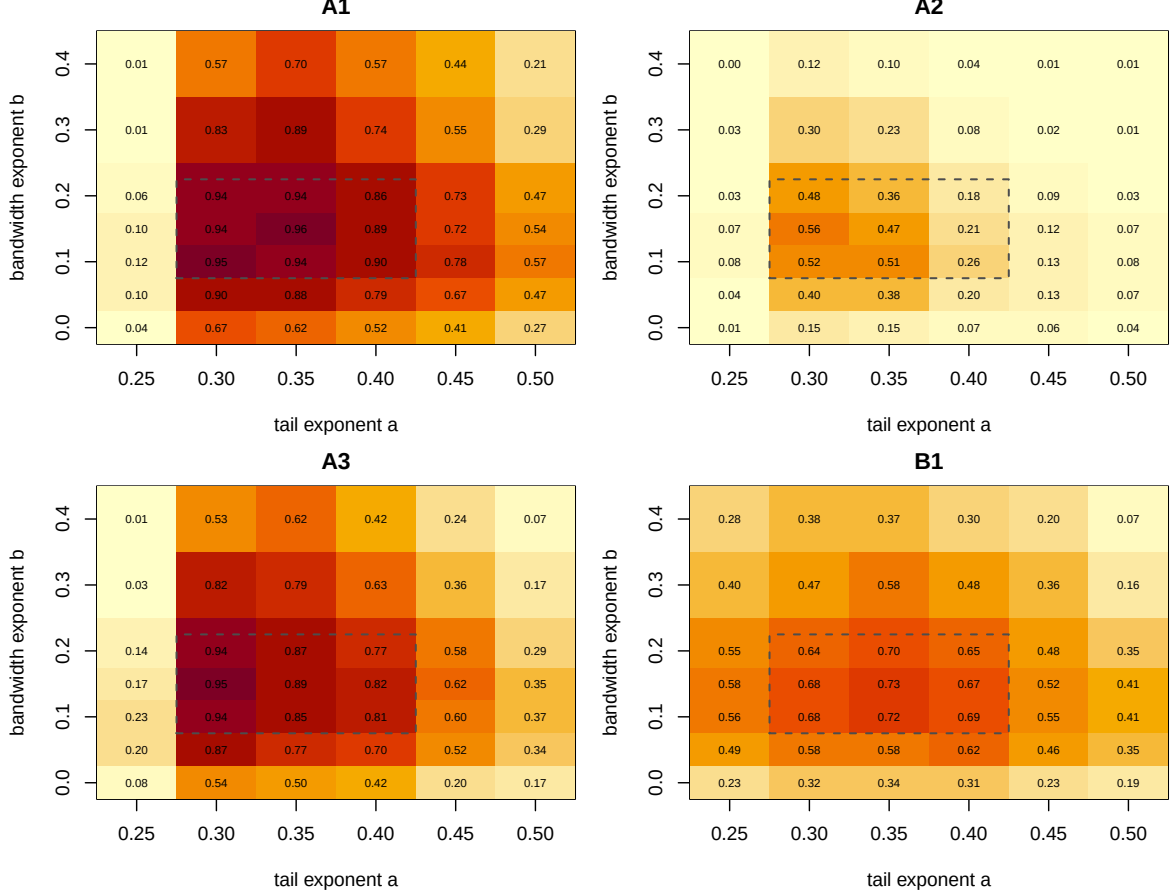


Figure 2: Sure-20 over the (a, b) tuning grid for the four models at $n = 2000$, $p = 1000$, $\rho = 0.25$ (200 common-random-number replications per model; $\alpha = n^{-a}$, $h = n^{-b}/2$; standard errors at most 0.036). The dashed rectangle marks the aggregation block $\mathcal{N}_9$.

The aggregated screen. A single pair (a, b) makes the final ranking depend entirely on one tuning. The aggregated version of the screen uses the 3×3 block of settings

$$\mathcal{N}_9 = \{0.30, 0.35, 0.40\} \times \{0.10, 0.15, 0.20\},$$

the 3×3 window anchored at (a^*, b^*) and extended away from the collapsed row $a = 0.25$. For each setting $(a, b) \in \mathcal{N}_9$ and each coordinate j , compute the score $\hat{\Psi}_j(a, b)$ and its rank $r_j(a, b)$ among the p coordinates, ranks increasing with the score; aggregate by

$$r_j^{\text{agg}} = \min_{(a, b) \in \mathcal{N}_9} r_j(a, b),$$

and rank the coordinates by increasing r_j^{agg} , ties broken by coordinate index. A coordinate is thus retained early whenever at least one reasonable tuning finds evidence for it. This direction of conservatism matches the purpose of a sure-screening first stage: false exclusion is final, while false retention is corrected by the second-stage multivariate analysis that the screen feeds. We make no optimality claim for the construction; it is the procedure used in the experiments and the application, and its effect is measured against the single-setting screen in Section 4.3. Its cost is nine score passes instead of one, leaving the $O(pn \log n)$ complexity unchanged up to the constant.

The two versions are reported side by side throughout: *Tail-index SIS, selected* (a^*, b^*) uses the central pair alone; *Tail-index SIS, aggregated* uses r_j^{agg} .

4.3 Comparison with competing screens

Six rules are compared on the same 1,000 simulated datasets per model and per dimension $p \in \{500, 1000, 2000\}$, at $n = 2000$, $\rho = 0.25$: the two versions of the tail-index screen; the conditional-Pickands screen of Yoshida and Umezu [19], implemented from their equations (2)–(4) with the Epanechnikov kernel and the tuning of their numerical work transplanted proportionally ($h = 1$, $k = \lfloor 0.072n \rfloor$), on 25 equally spaced evaluation points in $[\cdot 02, \cdot 98]$; and the quantile-adaptive SIS of He et al. [11] at the levels $\tau \in \{.90, .95, .99\}$, using cubic B-spline marginal quantile regression with three degrees of freedom, identical at every level. The competitors rank large utilities first; ours ranks small scores first. Tables 1–3 report, for each model and method, Sure- d at $d \in \{4, 20, 30, 50\}$ and the mean and median of the worst active rank; on B1, where the proxy block has size $q_{\text{scale}} = 20$, the values at $d = 30$ and $d = 50$ separate two different failure structures that Sure-20 alone conflates.

Sensitivity to the dimension. Every method degrades as p grows, at very different rates, and the mean–median contrast locates the degradation. For the selected-pair tail screen on A1, $\mathbb{E}(R_{\max})$ grows from 5.7 to 12.9 between $p = 500$ and $p = 2000$ while $\text{Med}(R_{\max})$ stays at 4: the typical replication is unaffected, and the loss is carried by a minority of replications in which an active coordinate falls into the null range, where its rank scales with p . Aggregation compresses that tail on every model ($\mathbb{E}(R_{\max})$ 4.6 $\rightarrow$ 6.1 on A1, 9.3 $\rightarrow$ 22.0 against 29.8 $\rightarrow$ 108.7 on B1, 4.9 $\rightarrow$ 7.9 against 7.0 $\rightarrow$ 22.1 on A3). The quantile screen at $\tau = .95$ is the least dimension-sensitive rule on the A family ($\mathbb{E}(R_{\max})$ 4.2 $\rightarrow$ 4.8 on A1). A2 is the most dimension-sensitive model for every method (aggregated Sure-20 .813 $\rightarrow$.580, quantile .95 .930 $\rightarrow$.782): a weak signal loses the most from added competition. Yoshida–Umezu degrades from a low base throughout ($\text{Med}(R_{\max})$ 29 $\rightarrow$ 168 on A1).

Sensitivity to the quantile level. The quantile screen is non-monotone in τ within every table, in opposite directions on either side of $\tau = .95$, and its behaviour on B1 tracks its target. At $\tau = .90$ its top four contain on average 3.8 of the twenty scale proxies, and all twenty fill its top 24; exact recovery is zero to three decimals at every p . This is coherent behaviour for a .90-quantile target, which the proxies genuinely move. The contamination is, however, *structured*: the active set is displaced behind the proxy block rather than lost, so Sure-30 jumps to .963–.990 and Sure-50 to .984–.997 — above the aggregated tail screen at the same d (.869–.934 and .919–.971). A practitioner who knew the size of the scale block could repair the quantile screen by retaining $d > q_{\text{scale}}$ variables; one who reads the leading positions, or must exactly recover, cannot. At $\tau = .95$ the picture is the same with smaller displacement (2.4–2.5 proxies in the top four, median worst active rank 20–21). At $\tau = .99$ the contamination largely clears (0.2–0.4 proxies in the top four) because the quantile target approaches the tail index, but the variance cost is severe everywhere: Sure-4 at $\tau = .95$ ranges from .234 (A2, $p = 2000$) to .891 (A1, $p = 500$) across the A family and falls to at most .125 at $\tau = .99$, and on B1 the level .99 never reaches Sure-4 .04. The direction of the τ -sensitivity does not depend on p ; its amplitude grows with p on the weak-signal A2. Yoshida–Umezu trails on all models and dimensions *as configured here*: its bandwidth $h = 1$ on the rank scale makes every evaluation point use nearly the whole sample, so its localization is weak, and our transplant of the tuning from their numerical work may not do it justice; a bandwidth sweep matched to the rank scale is left to future work, and no claim of dominance over a tuned version is made. It places 6.8–9.8 of the twenty B1 proxies in its top 24.

Table 1: Method comparison at $n = 2000$, $p = 500$, $\rho = 0.25$, on the same 1,000 replications per model. Sure- d is the probability that all four tail-index-active variables rank among the d smallest scores ($d = 4$ is exact recovery); note that B1 carries $q_{\text{scale}} = 20$ scale proxies, so $d = 20$ equals the block size and $d = 30, 50$ show the effect of retaining more than the block. $\mathbb{E}(R_{\text{max}})$ and $\text{Med}(R_{\text{max}})$ are the mean and median of the worst active rank. Standard errors of the probabilities are at most 0.016.

Model	Method	Sure-4	Sure-20	Sure-30	Sure-50	$\mathbb{E}(R_{\text{max}})$	$\text{Med}(R_{\text{max}})$
A1	Tail-index SIS, selected (a^*, b^*)	0.849	0.979	0.985	0.991	5.7	4
	Tail-index SIS, aggregated	0.922	0.990	0.994	0.998	4.6	4
	Yoshida–Umezū	0.202	0.467	0.510	0.569	97.4	29
	Quantile SIS ($\tau = .90$)	0.299	0.894	0.933	0.965	11.7	6
	Quantile SIS ($\tau = .95$)	0.891	0.998	1.000	1.000	4.2	4
	Quantile SIS ($\tau = .99$)	0.125	0.852	0.928	0.977	13.0	9
A2	Tail-index SIS, selected (a^*, b^*)	0.256	0.655	0.724	0.807	38.2	9
	Tail-index SIS, aggregated	0.245	0.813	0.864	0.914	19.5	7
	Yoshida–Umezū	0.028	0.178	0.226	0.305	153.9	124
	Quantile SIS ($\tau = .90$)	0.099	0.597	0.694	0.807	33.8	14
	Quantile SIS ($\tau = .95$)	0.431	0.930	0.962	0.988	8.2	5
	Quantile SIS ($\tau = .99$)	0.005	0.268	0.413	0.614	58.5	37
A3	Tail-index SIS, selected (a^*, b^*)	0.775	0.966	0.980	0.987	7.0	4
	Tail-index SIS, aggregated	0.839	0.986	0.992	0.997	4.9	4
	Yoshida–Umezū	0.181	0.509	0.557	0.626	79.8	20
	Quantile SIS ($\tau = .90$)	0.337	0.915	0.946	0.971	10.8	5
	Quantile SIS ($\tau = .95$)	0.863	0.996	0.996	0.999	4.4	4
	Quantile SIS ($\tau = .99$)	0.077	0.775	0.882	0.959	16.8	11
B1	Tail-index SIS, selected (a^*, b^*)	0.308	0.730	0.797	0.858	29.8	7
	Tail-index SIS, aggregated	0.523	0.911	0.934	0.971	9.3	4
	Yoshida–Umezū	0.007	0.088	0.175	0.285	132.0	108
	Quantile SIS ($\tau = .90$)	0.000	0.017	0.990	0.997	24.1	24
	Quantile SIS ($\tau = .95$)	0.030	0.514	0.994	1.000	18.0	20
	Quantile SIS ($\tau = .99$)	0.037	0.665	0.815	0.933	21.3	15

Effect of the aggregation. Aggregation improves the tail screen most where failures are worst. On B1 it lifts exact recovery from .308/.249/.234 to .523/.439/.402 across the three dimensions, divides $\mathbb{E}(R_{\text{max}})$ by three to five, and brings $\text{Med}(R_{\text{max}})$ to 4–5; on A1 and A3 it adds two to ten points of Sure-4 and compresses the rank tail; Sure-20 improves in all twelve model–dimension cells. On A1 and A3 the Sure-4 gains are 7.3/7.3/10.0 and 6.4/2.1/7.3 points across the three dimensions. On the weak-signal A2 aggregation trades a little exact recovery (.256 $\rightarrow$.245 at $p = 500$, .149 $\rightarrow$.113 at $p = 2000$) for substantially better inclusion (Sure-20 .463 $\rightarrow$.580 at $p = 2000$; $\mathbb{E}(R_{\text{max}})$ 144.8 $\rightarrow$ 76.0). The trade is the intended one: promotion by a single reasonable tuning occasionally admits a lucky null into the top four when the signal is weak, and reliably rescues active coordinates from single-setting failures further down the list. Aggregation stabilizes inclusion — the sure-screening objective — rather than exact selection.

The quantile screen at a moderate level remains the strongest detector when high quantiles and the limiting exponent share their covariates and the signal is weak (A2), and even under the B1

Table 2: Method comparison at $n = 2000$, $p = 1000$, $\rho = 0.25$, on the same 1,000 replications per model. Sure- d is the probability that all four tail-index-active variables rank among the d smallest scores ($d = 4$ is exact recovery); note that B1 carries $q_{\text{scale}} = 20$ scale proxies, so $d = 20$ equals the block size and $d = 30, 50$ show the effect of retaining more than the block. $\mathbb{E}(R_{\max})$ and $\text{Med}(R_{\max})$ are the mean and median of the worst active rank. Standard errors of the probabilities are at most 0.016.

Model	Method	Sure-4	Sure-20	Sure-30	Sure-50	$\mathbb{E}(R_{\max})$	$\text{Med}(R_{\max})$
A1	Tail-index SIS, selected (a^*, b^*)	0.809	0.954	0.965	0.979	8.3	4
	Tail-index SIS, aggregated	0.882	0.985	0.990	0.996	4.9	4
	Yoshida–Umezū	0.179	0.398	0.434	0.486	188.1	58
	Quantile SIS ($\tau = .90$)	0.219	0.804	0.872	0.930	18.6	7
	Quantile SIS ($\tau = .95$)	0.830	0.997	0.998	0.998	4.6	4
	Quantile SIS ($\tau = .99$)	0.068	0.674	0.812	0.928	22.2	15
A2	Tail-index SIS, selected (a^*, b^*)	0.192	0.525	0.589	0.672	81.0	17
	Tail-index SIS, aggregated	0.183	0.695	0.769	0.844	38.7	9
	Yoshida–Umezū	0.021	0.138	0.175	0.225	320.1	256
	Quantile SIS ($\tau = .90$)	0.059	0.456	0.557	0.683	68.7	24
	Quantile SIS ($\tau = .95$)	0.303	0.873	0.924	0.950	14.8	6
	Quantile SIS ($\tau = .99$)	0.004	0.103	0.197	0.350	120.5	74
A3	Tail-index SIS, selected (a^*, b^*)	0.747	0.938	0.949	0.967	11.0	4
	Tail-index SIS, aggregated	0.768	0.980	0.985	0.990	6.5	4
	Yoshida–Umezū	0.176	0.445	0.490	0.558	156.8	34
	Quantile SIS ($\tau = .90$)	0.249	0.845	0.900	0.930	18.3	7
	Quantile SIS ($\tau = .95$)	0.794	0.992	0.995	0.998	4.7	4
	Quantile SIS ($\tau = .99$)	0.040	0.569	0.716	0.857	28.4	18
B1	Tail-index SIS, selected (a^*, b^*)	0.249	0.654	0.715	0.782	59.9	9
	Tail-index SIS, aggregated	0.439	0.851	0.901	0.938	16.5	5
	Yoshida–Umezū	0.004	0.059	0.107	0.182	265.5	206
	Quantile SIS ($\tau = .90$)	0.000	0.014	0.970	0.984	25.7	24
	Quantile SIS ($\tau = .95$)	0.035	0.489	0.979	0.994	18.7	21
	Quantile SIS ($\tau = .99$)	0.020	0.449	0.619	0.812	35.2	23

cluster it recovers the active set at any budget exceeding the block size. The aggregated tail screen is competitive or better on A1 and A3, and on B1 it is the rule whose *leading* positions stay clean: at every d up to the block size it dominates, its top four contain 0.14–0.19 proxies against 2.1–3.8 for the quantile levels .95 and .90, and its median worst rank stays at 4–5 against 20–24. Which failure structure matters depends on the use: for exact recovery or for reading the leading variables, the tail screen; for inclusion at a generous budget known to exceed any scale block, a moderate-level quantile screen. For a question about the limiting exponent itself, the two rankings are best read side by side, as in Section 5: variables high on one list and low on the other are where the two targets part ways.

Table 3: Method comparison at $n = 2000$, $p = 2000$, $\rho = 0.25$, on the same 1,000 replications per model. Sure- d is the probability that all four tail-index-active variables rank among the d smallest scores ($d = 4$ is exact recovery); note that B1 carries $q_{\text{scale}} = 20$ scale proxies, so $d = 20$ equals the block size and $d = 30, 50$ show the effect of retaining more than the block. $\mathbb{E}(R_{\max})$ and $\text{Med}(R_{\max})$ are the mean and median of the worst active rank. Standard errors of the probabilities are at most 0.016.

Model	Method	Sure-4	Sure-20	Sure-30	Sure-50	$\mathbb{E}(R_{\max})$	$\text{Med}(R_{\max})$
A1	Tail-index SIS, selected (a^*, b^*)	0.751	0.941	0.952	0.967	12.9	4
	Tail-index SIS, aggregated	0.851	0.983	0.986	0.991	6.1	4
	Yoshida–Umezu	0.125	0.305	0.350	0.395	432.2	168
	Quantile SIS ($\tau = .90$)	0.161	0.665	0.769	0.843	36.6	11
	Quantile SIS ($\tau = .95$)	0.758	0.992	0.998	1.000	4.8	4
	Quantile SIS ($\tau = .99$)	0.025	0.406	0.574	0.769	40.2	25
A2	Tail-index SIS, selected (a^*, b^*)	0.149	0.463	0.517	0.609	144.8	27
	Tail-index SIS, aggregated	0.113	0.580	0.654	0.730	76.0	14
	Yoshida–Umezu	0.015	0.084	0.108	0.138	640.2	545
	Quantile SIS ($\tau = .90$)	0.039	0.328	0.425	0.544	122.2	42
	Quantile SIS ($\tau = .95$)	0.234	0.782	0.835	0.889	24.7	7
	Quantile SIS ($\tau = .99$)	0.000	0.022	0.058	0.147	245.7	154
A3	Tail-index SIS, selected (a^*, b^*)	0.650	0.892	0.912	0.935	22.0	4
	Tail-index SIS, aggregated	0.723	0.964	0.976	0.983	7.9	4
	Yoshida–Umezu	0.142	0.379	0.419	0.481	336.7	67
	Quantile SIS ($\tau = .90$)	0.171	0.723	0.787	0.853	36.4	9
	Quantile SIS ($\tau = .95$)	0.698	0.988	0.993	0.999	5.4	4
	Quantile SIS ($\tau = .99$)	0.023	0.328	0.462	0.667	56.2	34
B1	Tail-index SIS, selected (a^*, b^*)	0.234	0.599	0.640	0.720	108.7	11
	Tail-index SIS, aggregated	0.402	0.813	0.869	0.919	22.0	5
	Yoshida–Umezu	0.008	0.057	0.089	0.145	494.5	380
	Quantile SIS ($\tau = .90$)	0.000	0.012	0.963	0.992	25.1	24
	Quantile SIS ($\tau = .95$)	0.038	0.482	0.972	0.997	19.0	21
	Quantile SIS ($\tau = .99$)	0.008	0.254	0.383	0.584	61.7	39

5 Application: violent crime across U.S. communities

The analysis contains $n = 1,993$ communities and $p = 99$ candidate predictors. Under the tuning selected in Section 4.2, $\alpha = n^{-0.3} \approx 0.102$ and $h = n^{-0.15}/2 \approx 0.160$, so the sample contains $n\alpha \approx 204$ upper-tail observations in total and about $2nah \approx 65$ within a typical interior rank window. The dimension is therefore comparable to the number of observations available for local tail estimation. Unrestricted or nonparametric joint tail modelling with 99 covariates is poorly supported by roughly 200 upper-tail observations, and by fewer still within any local covariate neighbourhood; a marginal screen can rank the 99 candidates from exactly this information.

5.1 Data and protocol

The Communities and Crime data set [15], distributed by the UCI Machine Learning Repository in its unnormalized version, links socio-economic census attributes, law-enforcement survey attributes and FBI crime counts for 2,215 U.S. communities. The response is the violent-crime rate per 100,000 inhabitants. We keep the communities with an observed positive response ($n = 1,993$). Preprocessing of the 124 documented predictive attributes is response-blind: we retain numeric predictors with fewer than 5% missing values, which removes the 22 law-enforcement variables reported by only 343 communities, and with at least 100 distinct observed values, which removes 3 low-cardinality variables, keeping the application close to the continuous-marginal framework of the theory. This leaves $p = 99$ continuous predictors — demographic composition, income and poverty, education, employment, family structure, immigration, language, housing, and population density. The at most one missing value per retained predictor is median-imputed, and every method below works from the same imputed matrix.

The response is strongly right-skewed: median 374, 99th percentile 2,943, maximum 4,877. The left panel of Figure 3 shows the Hill curve of the response over $k \in [20, 600]$ upper order statistics; it is positive and moves slowly, between 0.283 and 0.374 for $k \in [100, 300]$, with value 0.328 at the working level $k = n\alpha = 204$. The middle panel shows the Pareto QQ plot of the 600 largest responses, approximately linear over the working range with least-squares slope 0.27. The gap between the QQ slope and the Hill value at the same level is mild evidence that regular variation is only approximate over the working range. The response exhibits a sufficiently heavy and stable upper tail to make a tail-index analysis plausible; neither diagnostic establishes regular variation beyond doubt, and no single threshold carries a structural interpretation.

The screening protocol is the one fixed in Section 4.2, applied unchanged: the aggregated tail-index screen computes the score at each of the nine settings $\mathcal{N}_9 = \{0.30, 0.35, 0.40\} \times \{0.10, 0.15, 0.20\}$, converts each pass into ranks, and orders the predictors by the minimum rank r_j^{agg} across the nine settings; the selected-pair screen at $(a^*, b^*) = (0.30, 0.15)$ is reported alongside. The two competing screens of Section 4.3 are run on the same imputed matrix: Yoshida–Umezū at its transplanted tuning and the quantile screen at $\tau \in \{.90, .95, .99\}$.

5.2 Results

Poverty and family structure dominate the aggregated ranking, and they dominate every competing ranking as well. The poverty rate (**pctPoverty**) is first, with per-setting ranks between 1 and 6 across the nine settings; the two-parent household variables **pct2Par** and **pctKids2Par** are third and fourth (ranks 1–5), and both are also leading variables for the quantile screen at every level (**pctKids2Par** is first at all three τ) and for Yoshida–Umezū (ranks 3–7). The related **pctKids4w2Par** and **pct12to17w2Par** complete the consensus block at aggregated ranks 7–8. The high-quantile and tail-index rankings share these leading predictors.

The informative disagreements are the tail-specific candidates. The share of workers using public transportation (**pctUsePubTrans**, aggregated rank 5) and the farm-employment rate (**pctWfarm**, rank 6) rank far lower under every fixed-quantile method: ranks 49–73 for the quantile screen across its three levels and 57–60 for Yoshida–Umezū. Their low tail-index scores are consistent with marginal variation in upper-tail heaviness across their values, of a kind that fixed-quantile screens are not designed to detect; Section 4.3 produces the same dissociation under controlled conditions, where the target is known. The two variables differ in stability across the aggregation block: **pctUsePubTrans** ranks within the top 15 at every one of the nine settings, while **pctWfarm** moves between 2 and 47 (right panel of Figure 3). The never-married male rate (**pctMaleNevMar**,

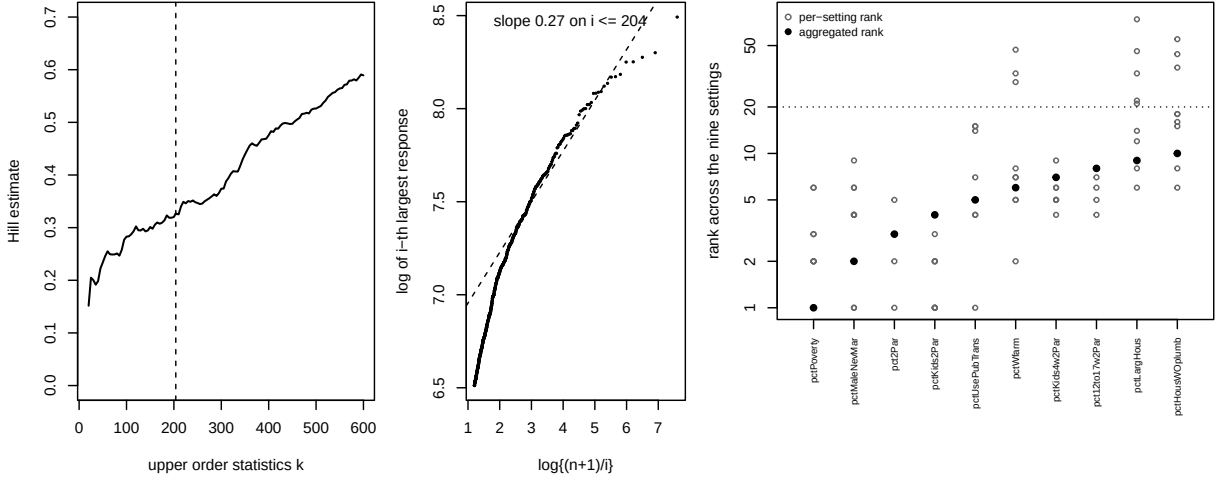


Figure 3: Communities and Crime application ($n = 1,993$, $p = 99$, $\alpha = n^{-0.3}$, $h = n^{-0.15}/2$). Left: Hill estimates for the violent-crime rate; the dashed line marks the working level $k = n\alpha = 204$. Middle: Pareto QQ plot of the 600 largest responses; the dashed line is fitted over the working range $i \leq 204$. Right: for the ten leading predictors of the aggregated screen, the nine per-setting ranks (open circles) and the aggregated rank (filled circles); the dotted line is the default retained size $d = 20$.

aggregated rank 2, per-setting ranks 1–9, quantile ranks 28–35) sits between the consensus block and the tail-specific pair. We report all three as candidate tail-specific predictors for a second-stage analysis.

Minimum-rank aggregation is deliberately conservative: a predictor is promoted whenever at least one tuning in the prespecified neighbourhood ranks it highly, which suits a first-stage sure screen, where false exclusion is more costly than retaining additional candidates. **pctLargHous** (aggregated rank 9) and **pctHousWOp1umb** (rank 10) illustrate the resulting behaviour: their positions rest on one favourable setting each (per-setting ranks 6–74 and 6–55; selected-pair ranks 46 and 36), so their rankings are markedly more tuning-sensitive than those of the consensus block or of the tail-specific candidates, which hold their positions across the whole block. The conservative screen retains them intentionally; the per-setting ranks describe this sensitivity, and discrimination among the retained predictors is left to the subsequent multivariate analysis.

We do not interpret any ranking causally: the screen identifies covariates along which the conditional tail index varies marginally, and confounding among the 99 predictors is certain. The ranks reported here carry no inferential calibration; Section 6 shows how a permutation threshold converts the screen into a test and what such a test refuses to declare. A second caution comes from the boundary experiment of Section 6: at $2nah \approx 65$ — the regime of this application, imposed by $n = 1,993$ — the tail screen ranked a pure scale cluster first on the language data, an artefact that disappeared at $2nah \approx 232$. The tail-specific candidates of this section (**pctUsePubTrans**, **pctWfarm**, **pctMaleNevMar**) could in principle be scale drivers promoted by the same second-order mechanism, and nothing in these data rules that out; they are candidates for a second-stage analysis under that explicit caveat, and grafting a known scale cluster onto this design, as in Section 6, would measure the contamination directly.

6 Application with known nuisance structure: token-surprisal tails of a language model

The crime application reads screening ranks without ground truth. This section builds a second application in which the status of a family of covariates is known — in one case exactly — so that the target separation demonstrated by model B1 can be observed on real data.

6.1 Response, data, and regime

For a document $w_1, \dots, w_T$ scored by a language model θ , let $S_t = -\log p_\theta(w_t \mid w_{<t})$ be the token surprisal and take

$$Y = \exp\left(\max_{t \leq T} S_t\right) = \left\{\min_{t \leq T} p_\theta(w_t \mid w_{<t})\right\}^{-1}.$$

If the conditional surprisal tails are exponential, $\mathbb{P}(S_t > s \mid x) \sim c(x)e^{-s/\gamma(x)}$ — the behaviour a Zipfian vocabulary produces — then each e^{S_t} is conditionally regularly varying with index $-1/\gamma(x)$, and for weakly dependent surprisals the maximum inherits the index: $\mathbb{P}(Y > y \mid x) \sim \theta(x) T c(x) y^{-1/\gamma(x)}$, where the extremal coefficient $\theta(x)$ and T enter the scale only. Weak dependence of the within-document surprisal series at extreme levels is an assumption of this application, not a verified fact; the nested-prefix experiment below supports its first-order consequence. Mean log-perplexity, by contrast, concentrates and lands in the Gumbel domain; its marginal can look heavy-tailed by mixture while every conditional law is light, which would invalidate (C1) silently. The maximum, not the mean, is the object this framework applies to.

The corpus is a 2025 Common Crawl snapshot (FineWeb), chosen because its filtering is heuristic: corpora filtered by perplexity or by quality classifiers truncate the upper tail of Y , a selection on the response. The model is Pythia-410m (fp32 throughout; the reference model of the likelihood-ratio variant is Pythia-1.4b), whose public training corpus predates the crawl. The primary analysis scores exactly $T = 512$ tokens per document after a 64-token burn-in, so document length is fixed by construction; a secondary analysis leaves T free. Covariates are never computed from the forward pass that produces Y : $p = 272$ surface and frozen-encoder-embedding covariates survive a continuity filter, with $n = 20,000$ documents, and at the tuning of Section 4.2 the regime is $2nah \approx 232$ local extremes per interior window. The marginal generalized Pareto fit gives $\xi = 1.16$ with profile confidence interval $[1.07, 1.25]$, the marginal Hill estimate at the 10% level is 1.28 — the gap between the two reflecting threshold sensitivity — and the sixteen stratified fits give $\xi \in [1.14, 1.29]$ with every profile interval excluding 0: the conditional positivity that (C1) requires and that a marginal Hill plot cannot establish. Numerical saturation, degenerate document classes, and fp32 rounding are ruled out by direct checks in the replication material.

6.2 Calibrated screening and a null result

Ranks alone never refuse to answer: a ranking exists whether or not any signal does. The screening here is therefore calibrated by permutation: for $B = 1,000$ permutations σ of the responses against the covariate matrix, the full screen is rerun and the familywise threshold is the empirical 5% quantile of the null minimum score $\min_j \widehat{\Psi}_j^{(\sigma)}$ (scores rank small-first). The null hypothesis this tests is joint independence of Y and the covariates — permutation destroys scale dependence and index dependence alike — so it is a global test, not a test of the tail-index-specific hypothesis; a covariate could move the scale of Y and reject under it. On the primary analysis, no covariate of the 272 crosses the familywise threshold, for either response variant. The per-covariate 5% thresholds are exceeded 31 times for Y and 23 for Y_Δ against 13.6 expected under independence, an excess

concentrated in the strongly dependent embedding-PCA block and absorbed by the familywise threshold; we read the outcome as no familywise-significant covariate, not as a textbook uniform null. The same holds for 1,024-dimensional blocks of internal model activations (residual-stream averages over a past prefix, screened against Y computed on the disjoint future suffix, at $n = 50,000$): every layer configuration is at or near chance level. An uncalibrated reading of the raw rankings would have produced discussable leaders in every one of these experiments; the calibration is what makes the null legible, and we regard it as part of the method’s proper use.

6.3 Two scale clusters with known status

Document length: a natural nuisance, asymptotic truth. For weakly dependent regularly varying surprisals, $\mathbb{E}[\max_{t \leq T} S_t \mid x] = \gamma(x) \log T + O(1)$: length enters the scale of Y and never its index. On nested prefixes of the same documents — composition held fixed — the measured slope of $\mathbb{E}[\max S]$ against $\log T$ is 1.23 with intra-document confidence interval [1.15, 1.31], containing the independent Hill estimates of γ ; across free-length strata the Hill intervals overlap while the location moves by 2.24 nats. When T is left free, the screening runs on the 16 surface covariates that survive the continuity filter in that mode, five of which are length proxies (characters, bytes, words, whitespace count, gzip size). Those five fill the top five positions of the quantile screen at every level $\tau \in \{.90, .95, .99\}$ — for one level, the chance probability that a prespecified five-element block occupies the top five of sixteen is $1/\binom{16}{5} \approx 2.3 \times 10^{-4}$, and the three levels are strongly dependent, so we quote no combined probability — while the tail-index screen places none of them in its top five and ranks the block in its bottom quartile (median rank 13 of 16 against 3 for the quantile screen). Under the permutation calibration of the free- T run, no covariate reaches the familywise threshold for Y , and a single covariate crosses for Y_Δ . This is the B1 dissociation on real data, with a nuisance an order of magnitude stronger than the simulated one, visible in the rankings and absorbed by the familywise test.

The same experiment locates a boundary of validity: at $n = 2,000$ ($2nah \approx 65$) the tail screen itself ranks the length block first — the second-order bias of the local Hill statistic responds to a scale shift of this amplitude — and the artefact disappears at $n = 20,000$ ($2nah \approx 232$). Specificity against scale is an asymptotic property with a measurable finite-sample onset.

A grafted cluster: exact truth. By Breiman’s lemma, if $\mathbb{P}(Y > y \mid x)$ is regularly varying with index $-1/\gamma(x)$ and $V > 0$ is bounded and independent of Y given x , then $\mathbb{P}(YV > y \mid x) \sim \mathbb{E}\{V^{1/\gamma(x)}\} \mathbb{P}(Y > y \mid x)$: the index is unchanged and only the scale constant moves (boundedness makes the moment finite for every γ). Grafting $V = \exp\{\kappa(\Phi(F) - 1/2)\}$ with a latent $F \sim N(0, 1)$ and $q_{\text{scale}} = 20$ observed proxies of F (loading 0.7, as in B1) onto the real responses produces twenty covariates that provably carry no information about the index while shifting every finite quantile, inside the real correlation structure of the 272 genuine covariates. Across the paired strength sweep $\kappa \in \{0, 0.5, 1, 2, 4\}$, the number of proxies among the twenty leading positions grows for the quantile screen from 1 ($\kappa = .5$) to 19 ($\kappa = 2$) and 20 ($\kappa = 4$, all three levels; median proxy rank 10.5 of 292). The tail-index screen retains 2 to 4 (selected pair; median proxy rank 143–175, at or beyond the chance value of 146) and 0 to 5 (aggregated; median proxy rank 126–181 up to $\kappa = 2$, falling to 66.8 at $\kappa = 4$). The quantile screen’s behaviour is coherent for its own target throughout; the experiment measures an order-of-magnitude sensitivity gap, not an immunity — at $\kappa = 4$ both variants of the tail screen show visible contamination (4–5 proxies, one of them first), and minimum-rank aggregation does not help here and degrades the median proxy rank, since there is no dispersed tail signal to stabilize and nine passes only pool noise. The two findings must be kept separate: under the permutation calibration nothing is declared in any arm for any screen — no proxy ever reaches

the familywise threshold, so neither method makes a false discovery — while the target separation of this section lives in the uncalibrated rankings, which are what a practitioner reads when choosing d variables to retain. The raw rankings are more contaminated than the calibrated procedure.

6.4 What this application does and does not establish

An external criterion, computed without any screen — the dispersion of stratified γ estimates per covariate, calibrated against random strata — identifies which covariates actually modulate the index in these data (a single clear case, the whitespace rate, with non-overlapping quartile intervals). Both screening families rank γ -modulating covariates equally well by this criterion, the quantile screen at $\tau = .95$ marginally better. The evidence of this section is therefore evidence of *specificity*: covariates that move finite quantiles without touching the index — whether by asymptotic argument (length) or by exact construction (the grafted cluster) — capture every fixed-quantile ranking and not the tail-index ranking. It is not evidence of superior power for finding index-active covariates, and the simulations of Section 4.3 say the same: the advantage of the tail-index screen is what its leading positions contain, not how often it finds a shared signal first.

7 Discussion

The screen is a first stage. Its intended pipeline reduces p coordinates to a manageable d , after which a joint method — the central tail-index subspace of Gardes and Podgorny [9], a structured tail regression [17, 18], or multivariate local estimation — can resolve interactions and correlated neighbours among the retained variables. The two stages solve different problems: rapid elimination of evidently irrelevant coordinates scales to $p \gg n$, while accurate joint tail inference does not.

Three limitations are intrinsic. First, activity without marginal detectability: when every fibre of an active coordinate meets the global argmax set, the score carries no information about it, at any sample size. Proposition 2.2 characterizes this failure exactly, and it is a property of the target, so enlarging d or n cannot repair it; iterative or grouped extensions would require new identification arguments. Second, mixture bias: when the projected slowly varying factor decays logarithmically, condition (E2) restricts the detectable gaps to $\Delta_{\min} \gg 1/\log n$, and the continuous-mixture example of Section 3 shows the restriction is intrinsic rather than an artifact of the proofs. Third, rare favourable configurations: even with a comfortable population gap, the envelope near its maximum is estimable only from observations whose remaining active coordinates are simultaneously near their maximizing values, and the mass of such configurations can decay exponentially with the number of active coordinates. The weak-signal model A2 in Section 4 shows the resulting cost.

The main open theoretical question is the gap between the primitive model (C1)–(C2) and the estimation conditions (E1)–(E3). Projection creates slow variation that the primitive model does not control, and we do not know useful primitive conditions implying uniform triangular-array moduli for all coordinatewise projections. The score functional also invites variants: a minimum or a low quantile of the profile would emphasize depressions confined to short intervals, at the price of new uniformity arguments for empirical profile extrema.

In practice, tuning matters more than the theory prescribes. The tail fraction trades Hill bias against variance; the bandwidth additionally determines how often near-maximizing configurations of the other active coordinates enter a window. We recommend the workflow of Sections 4 and 5: run the screen on the block of nine neighbouring tunings and rank by the aggregated minimum rank. The aggregation is deliberately conservative — a predictor is retained when at least one reasonable tuning finds evidence for it — which is the right direction of error for a first-stage sure screen: false exclusion is final, while false retention is corrected by the second-stage multivariate analysis. The

per-setting ranks describe tuning sensitivity and should accompany the ranking as a diagnostic, not act as a second screening threshold. Dependence deserves caution of its own: it can help, by concentrating active configurations, or hurt, by promoting inactive neighbours, and the direction is design-specific.

Projection turns a tail-index surface into an upper envelope. Variation of that envelope is a scalable marginal signal for tail-index activity, and its absence, characterized by the argmax projections, tells the analyst exactly what a marginal screen can and cannot see.

A Auxiliary results

This appendix collects the technical tools used by the proofs of Appendix B: elementary probabilistic lemmas (A.1), the projection lemmas behind Proposition 2.1 (A.2), the deterministic geometry of empirical-rank windows (A.3), the local Hill representation and the finite-sample uniform profile bound (A.4), and the passage from profiles to scores, ending with a deterministic separation lemma (A.5).

Throughout, the dimension is written $p = p_n$, and two auxiliary moduli quantify the content of (E2)–(E3):

$$b_n(a) = \max_{j \leq p} \sup_{u \in I_\varepsilon} \sup_{s \geq a^{-1}, t > 1} \frac{|\log\{\ell_j(ts, u)/\ell_j(s, u)\}|}{\log t}, \quad \omega_n(r, \tau) = \max_{j \leq p} \sup_{u \in I_\varepsilon} \sup_{\substack{v \in [0, 1] \\ |v - u| \leq r}} \sup_{w \in [\tau, 1 - \tau]} \left| \log \frac{Q_j(w^{-1}, v)}{Q_j(w^{-1}, u)} \right|,$$

so that (E2) reads $b_n(3\alpha) = o(\Delta_{\min})$ and (E3) reads $\omega_n(C_0 h, \tau_n) = o(\Delta_{\min})$ with $\tau_n = (pn)^{-2}$; $\omega_n(\cdot, \tau)$ is nondecreasing in its first argument, so (E3) bounds $\omega_n(r, \tau_n)$ for every $r \leq C_0 h$. Put $\delta_n = (n + 1)^{-1}$,

$$m_n^- = \lfloor \{h + \min(h, \varepsilon)\}(n + 1) \rfloor - 1, \quad K_n = \lfloor \alpha m_n^- \rfloor, \quad S_n = 4n + 1.$$

All conditional distributions are the pointwise compatible versions of Section 2, and constants are uniform over the triangular array.

A.1 Elementary probabilistic lemmas

The first lemma transfers pointwise perturbations to order statistics, as in Gardes and Podgorny [9, Appendix, Lemma 3].

Lemma A.1 (Sorting is sup-norm nonexpansive). *Let $a_1, \dots, a_m$ and $b_1, \dots, b_m$ satisfy $a_i \leq b_i$ for all i . Then $a_{(i)} \leq b_{(i)}$ for all i . Consequently $|a_i - b_i| \leq \epsilon$ for all i implies $|a_{(i)} - b_{(i)}| \leq \epsilon$ for all i .*

Proof. Fix i . At least $m - i + 1$ indices r satisfy $a_r \geq a_{(i)}$, and for each such r , $b_r \geq a_r \geq a_{(i)}$; hence $b_{(i)} \geq a_{(i)}$. For the second claim apply the first to $(a_i - \epsilon, b_i)$ and $(b_i - \epsilon, a_i)$. $\square$

Lemma A.2 (Chernoff bound for exponential means). *Let L_k be the mean of k i.i.d. standard exponentials. For $0 < t \leq 1$, $\mathbb{P}(|L_k - 1| > t) \leq 2e^{-kt^2/4}$.*

Proof. Markov's inequality applied to $e^{\lambda k L_k}$, optimized at $\lambda = t/(1 + t)$, gives $\mathbb{P}(L_k > 1 + t) \leq e^{-k\{t - \log(1 + t)\}}$. The function $\varphi(t) = t - \log(1 + t) - t^2/4$ has $\varphi(0) = 0$ and

$$\varphi'(t) = 1 - \frac{1}{1 + t} - \frac{t}{2} = \frac{t(1 - t)}{2(1 + t)} \geq 0 \quad \text{on } [0, 1],$$

so $t - \log(1 + t) \geq t^2/4$ on $(0, 1]$. For the lower tail, $\mathbb{P}(L_k < 1 - t) \leq e^{-k\{-t - \log(1 - t)\}}$ and $-t - \log(1 - t) \geq t^2/2 \geq t^2/4$. $\square$

Lemma A.3 (Threshold order statistic of uniforms). *Let $V_{(1)} < \dots < V_{(m)}$ be the order statistics of m i.i.d. standard uniforms, $0 < \alpha \leq 1/3$ and $k = \lfloor \alpha m \rfloor \geq 1$. Then $\mathbb{P}\{V_{(k+1)} > 3\alpha\} \leq e^{-(k+1)/4}$.*

Proof. If $3\alpha \geq 1$ the probability is zero. Otherwise $\{V_{(k+1)} > 3\alpha\} = \{N \leq k\}$ with $N \sim \text{Bin}(m, 3\alpha)$ and mean $\mu = 3\alpha m \geq 3k$. The multiplicative Chernoff bound $\mathbb{P}\{N \leq (1 - \delta)\mu\} \leq e^{-\delta^2\mu/2}$ [3, Section 2.2] with $\delta = 2/3$ gives $\mathbb{P}(N \leq \mu/3) \leq e^{-2\mu/9} \leq e^{-2k/3} \leq e^{-(k+1)/4}$, the last step because $8k \geq 3(k+1)$ for $k \geq 1$. $\square$

A.2 Projection lemmas

Let $\mathcal{G} \subseteq \sigma(U)$, $T = \gamma(U)$, and $\xi = \text{ess sup}(T \mid \mathcal{G})$, the smallest $\mathcal{G}$ -measurable random variable dominating T almost surely.

Lemma A.4 (Uniform remainder replacement). *Define $M(y) = \mathbb{E}\{c(U)y^{-1/T} \mid \mathcal{G}\}$. Under (C1)–(C2), almost surely,*

$$\left| \frac{\mathbb{P}(Y > y \mid \mathcal{G})}{M(y)} - 1 \right| \leq \rho(y) := \sup_u \left| \frac{L(y, u)}{c(u)} - 1 \right| \longrightarrow 0. \quad (\text{A.1})$$

Proof. By the tower property, $\mathbb{P}(Y > y \mid \mathcal{G}) = \mathbb{E}\{y^{-1/T} L(y, U) \mid \mathcal{G}\}$. Writing $L(y, U) = c(U)\{1 + r(y, U)\}$ with $|r(y, U)| \leq \rho(y)$,

$$|\mathbb{P}(Y > y \mid \mathcal{G}) - M(y)| = |\mathbb{E}\{c(U)y^{-1/T} r(y, U) \mid \mathcal{G}\}| \leq \rho(y)M(y),$$

and $\rho(y) \rightarrow 0$ is (C2). $\square$

The next lemma is the point at which the argument departs from the proof of Proposition 1 in Gardes and Podgorny [9]: their condition (C.1) assumes the law of $\gamma(X)$ absolutely continuous, which identifies $\{\gamma > t\}$ with $\{\gamma \geq t\}$ in their computation. The tilting argument below needs no such assumption and allows purely atomic conditional laws of T — the relevant case for inactive coordinates and constant-index null models.

Lemma A.5 (Tilted-measure concentration). *For $y > 1$ define the random probability measure*

$$K_y^\star(A) = \frac{\mathbb{E}\{c(U)y^{-1/T} \mathbf{1}_{\{T \in A\}} \mid \mathcal{G}\}}{M(y)}, \quad A \in \mathcal{B}([\underline{\gamma}, \Gamma]).$$

On a common probability-one event, $K_y^\star \Rightarrow \delta_\xi$ weakly as $y \rightarrow \infty$.

Proof. Since $K_y^\star\{T > \xi + \delta\} = 0$ for every $\delta > 0$ by definition of ξ , it suffices to prove, for fixed $\delta \in (0, \underline{\gamma})$,

$$K_y^\star\{T \leq \xi - \delta\} \longrightarrow 0 \quad \text{almost surely.} \quad (\text{A.2})$$

Split by the $\mathcal{G}$ -measurable event $G_\delta = \{\xi \geq \underline{\gamma} + \delta\}$. On G_δ^c , $\xi - \delta < \underline{\gamma} \leq T$, so $\{T \leq \xi - \delta\}$ is empty and (A.2) is trivial. On G_δ , set $r = \xi - \delta$, $s = \xi - \delta/2$ and $\pi = \mathbb{P}(T > s \mid \mathcal{G})$. Then $\pi > 0$ almost surely on G_δ : otherwise $\xi' = \xi - (\delta/2)\mathbf{1}_{G_\delta \cap \{\pi=0\}}$ would be a $\mathcal{G}$ -measurable almost-sure upper bound for T , strictly smaller than ξ with positive probability, contradicting minimality of the conditional essential supremum. Bounding numerator and denominator separately, using only $c_- \leq c \leq c_+$ and monotonicity of $g \mapsto y^{-1/g}$,

$$K_y^\star\{T \leq r\} \leq \frac{c_+ y^{-1/r}}{c_- \pi y^{-1/s}} = \frac{c_+}{c_- \pi} y^{-(1/r - 1/s)} \longrightarrow 0.$$

A rational sequence $\delta \downarrow 0$ and a countable intersection give a common probability-one event. $\square$

Lemma A.6 (Continuity of the envelope). *Under (C1), the map $u \mapsto \max_v \gamma(u, v)$ is continuous on $[0, 1]$, with modulus bounded by the modulus of continuity of γ ; under (S) this map equals ξ_j on I_ε , by Proposition 2.1.*

Proof. γ is uniformly continuous on the compact cube; if $|u - u'| \leq \delta$ then $|\gamma(u, v) - \gamma(u', v)| \leq \varpi(\delta)$ for every v , and taking maxima preserves the bound. $\square$

Lemma A.7 (Regular variation of the projected quantile). *If $\bar{F}(y) = \mathbb{P}(Y > y \mid U_j = u)$ is regularly varying at infinity with index $-1/\xi$, $\xi > 0$, then its generalized upper quantile $Q(s) = \bar{F}^{-1}(1/s)$ is regularly varying with index ξ , i.e. $Q(s) = s^\xi \ell(s)$ with ℓ slowly varying.*

Proof. This is the standard inversion property of regularly varying functions [6, Proposition B.1.9(9)]: the generalized inverse of a regularly varying nonincreasing function with index $-1/\xi$ is regularly varying with index ξ at the appropriate argument, and the representation follows from the definition of slow variation. $\square$

A.3 Empirical-rank geometry

The three lemmas of this subsection are purely deterministic or rank-based; no property of the response enters. They replace the covariate-density conditions (H.1)–(H.2) of Gardes and Podgorny [9] in the present empirical-rank setting.

Lemma A.8 (Rank displacement). *Let $\hat{U}_{ij} = R_{ij}/(n+1)$ and*

$$D_n(\eta) = \left\{ \max_{j \leq p} \max_{i \leq n} |U_{ij} - \hat{U}_{ij}| \leq \eta + \delta_n \right\}.$$

Then $\mathbb{P}\{D_n(\eta)^c\} \leq 2pe^{-2n\eta^2}$.

Proof. Fix j and let $F_{n,j}$ be the empirical distribution function of the U_{ij} , $i \leq n$. By the Dvoretzky–Kiefer–Wolfowitz inequality with Massart’s constant [14], $\sup_t |F_{n,j}(t) - t| \leq \eta$ outside probability $2e^{-2n\eta^2}$. On that event, continuity gives $F_{n,j}(U_{(r)j}) = r/n$, hence $|U_{(r)j} - r/n| \leq \eta$ and

$$|U_{(r)j} - r/(n+1)| \leq \eta + \frac{r}{n(n+1)} \leq \eta + \delta_n$$

for every rank r simultaneously. A union bound over j finishes; no union over ranks is needed. $\square$

Lemma A.9 (Deterministic window counts). *For every j , every realization and every $u \in I_\varepsilon$, $M_j(u) \geq m_n^-$ and $k_j(u) \geq K_n$; moreover $M_j(u)$ is a nonrandom function of u .*

Proof. The ranks form a permutation of $1, \dots, n$, so $M_j(u) = \#\{r \in \{1, \dots, n\} : |r/(n+1) - u| \leq h\}$, a deterministic function of u : the number of integers in $[x(u), y(u)]$ with $x(u) = \max\{(u-h)(n+1), 1\}$ and $y(u) = \min\{(u+h)(n+1), n\}$. Fix $u \in I_\varepsilon$ and write $m^- = m_n^- = \lfloor \{h + \min(h, \varepsilon)\}(n+1) \rfloor - 1$; note $h + \min(h, \varepsilon) \leq \min(2h, h + \varepsilon)$. Four cases.

If neither endpoint is clipped, the interval has length $2h(n+1)$ and contains at least $\lfloor 2h(n+1) \rfloor - 1 \geq m^-$ integers.

If only the left endpoint is clipped, then $x(u) = 1$ and the count is $\lfloor y(u) \rfloor$. Either $y(u) = (u+h)(n+1) \geq (\varepsilon+h)(n+1)$, so the count is at least $\lfloor (h+\varepsilon)(n+1) \rfloor \geq m^- + 1$; or $y(u) = n$ and the count is $n \geq \lfloor (h+\varepsilon)(n+1) \rfloor - 1 \geq m^-$, using $h \leq 1 - \varepsilon$, i.e. $(h+\varepsilon)(n+1) \leq n+1$.

If only the right endpoint is clipped, then $y(u) = n$ and the count is $n - \lceil x(u) \rceil + 1 \geq n - x(u) \geq n - (1 - \varepsilon - h)(n+1) = (h+\varepsilon)(n+1) - 1$, hence at least $\lceil (h+\varepsilon)(n+1) \rceil - 1 \geq m^-$.

If both endpoints are clipped, the count is n , which is at least m^- as in the second case. In every case $M_j(u) \geq m^-$, and $k_j(u) = \lfloor \alpha M_j(u) \rfloor \geq \lfloor \alpha m_n^- \rfloor = K_n$. $\square$

Lemma A.10 (Finite-state structure). *For fixed j and fixed ranks, $u \mapsto \mathcal{W}_j(u)$ takes at most $S_n = 4n + 1$ distinct values on $[0, 1]$.*

Proof. The membership indicator of observation i changes only at $\hat{U}_{ij} \pm h$: at most $2n$ breakpoints, hence at most $2n + 1$ constancy intervals plus the $2n$ breakpoint states. $\square$

Lemma A.11 (Local extreme count). *If $nh \geq 4$ and $\alpha m_n^- \geq 2$, then $m_n^- \geq nh/2$ and*

$$\frac{n\alpha h}{4} \leq K_n \leq 2n\alpha h + 1.$$

Proof. $m_n^- \geq \lfloor h(n+1) \rfloor - 1 \geq hn - 2 \geq nh/2$ for $nh \geq 4$, and $K_n = \lfloor \alpha m_n^- \rfloor \geq \alpha m_n^-/2 \geq n\alpha h/4$ when $\alpha m_n^- \geq 2$ (since $\lfloor x \rfloor \geq x/2$ for $x \geq 2$). For the upper bound, $m_n^- \leq 2h(n+1) - 1 \leq 2hn + 2h - 1 \leq 2hn + 1$, so $K_n \leq \alpha m_n^- \leq 2n\alpha h + 1$. $\square$

The rate conditions (7) force $nh \rightarrow \infty$ and $K_n \rightarrow \infty$, so Lemma A.11 applies for n large; it is the only conversion between K_n and $n\alpha h$ used in Appendix B.

A.4 Local Hill representation and uniform profile control

The scheme parallels the proof of Theorem 3 in Gardes and Podgorny [9] — oscillation control, exponential-spacings representation, slowly varying remainder — with the covering net over their compact matrix class replaced by exact union bounds over p coordinates and S_n window states.

Lemma A.12 (Conditional PIT and Rényi representation). *Fix j and condition on the column $\mathbf{U}_j = (U_{1j}, \dots, U_{nj})$. Under (E1):*

1. $V_{ij} = 1 - F_{Y|j}(Y_i | U_{ij})$, $i \leq n$, are i.i.d. standard uniform, and $Y_i = Q_j(1/V_{ij}, U_{ij})$ almost surely;
2. for any index set $\mathcal{W}$ of size m , deterministic given $\mathbf{U}_j$, with $V_{(1)} < \dots < V_{(m)}$ the ordered PIT values of its members and $k = \lfloor \alpha m \rfloor$,

$$L_k = \frac{1}{k} \sum_{r=1}^k \log \frac{V_{(k+1)}}{V_{(r)}}$$

is distributed as the mean of k i.i.d. standard exponentials, independent of $V_{(k+1)}$.

Proof. (i) Given $\mathbf{U}_j$, the pairs (U_{ij}, Y_i) are independent in i and Y_i has continuous conditional distribution function $F_{Y|j}(\cdot | U_{ij})$, so V_{ij} is standard uniform. For the representation, $Q_j(1/V_{ij}, U_{ij}) = F_{Y|j}^{-1}(F_{Y|j}(Y_i | U_{ij}) | U_{ij}) \leq Y_i$ always, with strict inequality only when Y_i lies in the interior of an interval on which $F_{Y|j}(\cdot | U_{ij})$ is constant; each such interval has conditional probability zero and there are countably many, so the representation holds almost surely. Strict monotonicity is not needed. Conditioning on the whole column, rather than on window counts as in Gardes and Podgorny [9, Appendix, Lemma 4], makes the window sets deterministic (Lemma A.9); only coordinate j is conditioned on, and no independence across coordinates is used.

(ii) With $W_i = -\log V_i$ standard exponentials and order statistics $W_{(1)} < \dots < W_{(m)}$, $W_{(m-r+1)} = -\log V_{(r)}$, Abel summation gives

$$kL_k = \sum_{r=1}^k \{W_{(m-r+1)} - W_{(m-k)}\} = \sum_{i=1}^k i \{W_{(m-i+1)} - W_{(m-i)}\},$$

and by the Rényi representation of exponential order statistics [16] the normalized top spacings $i\{W_{(m-i+1)} - W_{(m-i)}\}$, $i \leq k$, are i.i.d. standard exponential and independent of $W_{(m-k)}$, hence of $V_{(k+1)}$. $\square$

Lemma A.13 (Frozen-location Hill expansion). *Fix j , condition on $\mathbf{U}_j$, and fix a window state $\mathcal{W}$ of size m realized at $u \in I_\varepsilon$, with $k = \lfloor \alpha m \rfloor$ and Hill statistic H . On the event $D_n(\eta)$ intersected with $\{V_{ij} \in [\tau, 1 - \tau] \ \forall i\}$ and $\{V_{(k+1)} \leq 3\alpha\}$,*

$$|H - \xi_j(u)| \leq 2\omega_n(h + \eta + \delta_n, \tau) + \xi_j(u)|L_k - 1| + b_n(3\alpha)L_k. \quad (\text{A.3})$$

Proof. Spatial freezing. Each $i \in \mathcal{W}$ has $|\widehat{U}_{ij} - u| \leq h$, hence $|U_{ij} - u| \leq r_n := h + \eta + \delta_n$ on $D_n(\eta)$. With $V_{ij} \in [\tau, 1 - \tau]$, the definition of ω_n gives $|\log Q_j(1/V_{ij}, U_{ij}) - \log Q_j(1/V_{ij}, u)| \leq \omega_n(r_n, \tau)$ for every member. Let $\check{H}$ be the Hill statistic of the frozen responses $\check{Y}_i = Q_j(1/V_{ij}, u)$. By Lemma A.1, ordered log-responses differ by at most $\omega_n(r_n, \tau)$, and the Hill statistic is an average of differences of two of them, so $|H - \check{H}| \leq 2\omega_n(r_n, \tau)$.

Exact decomposition. Since $s \mapsto Q_j(s, u)$ is nondecreasing, $\check{Y}_{(m-r+1)} = Q_j(1/V_{(r)}, u)$, and $\log Q_j(s, u) = \xi_j(u) \log s + \log \ell_j(s, u)$ yields

$$\check{H} = \xi_j(u)L_k + R, \quad R = \frac{1}{k} \sum_{r=1}^k \log \frac{\ell_j(1/V_{(r)}, u)}{\ell_j(1/V_{(k+1)}, u)}.$$

Remainder. On $\{V_{(k+1)} \leq 3\alpha\}$, each ratio is $\ell_j(ts, u)/\ell_j(s, u)$ with $s = 1/V_{(k+1)} \geq (3\alpha)^{-1}$ and $t = V_{(k+1)}/V_{(r)} > 1$, so $|\log(\cdot)| \leq b_n(3\alpha) \log(V_{(k+1)}/V_{(r)})$; averaging gives $|R| \leq b_n(3\alpha)L_k$. $\square$

The decomposition behind (A.3) is the shape of the whole argument:

$$\widehat{\xi}_j(u) - \xi_j(u) = \text{rank/spatial error} + \text{exponential fluctuation} + \text{tail remainder}.$$

Proposition A.14 (Finite-sample uniform profile bound). *Assume (C1)–(C2), (S), the structural part of (E1) (continuity of every $F_{Y|j}(\cdot | u)$, hence the quantile representation), that $b_n(3\alpha)$ and $\omega_n(\cdot, \cdot)$ are finite, $0 < h \leq 1 - \varepsilon$, $0 < \alpha \leq 1/3$, and $K_n \geq 2$. For every $\eta > 0$, $0 < t \leq 1$, $0 < \tau < 1/2$,*

$$\begin{aligned} \mathbb{P} \left\{ \max_{j \leq p} \sup_{u \in I_\varepsilon} |\widehat{\xi}_j(u) - \xi_j(u)| > 2\omega_n(h + \eta + \delta_n, \tau) + \Gamma t + (1 + t)b_n(3\alpha) \right\} \\ \leq \underbrace{2pe^{-2n\eta^2}}_{\text{ranks}} + \underbrace{2pn\tau}_{\text{PIT truncation}} + \underbrace{2pS_n e^{-K_n t^2/4}}_{\text{spacings}} + \underbrace{pS_n e^{-(K_n+1)/4}}_{\text{threshold}}. \end{aligned} \quad (\text{A.4})$$

Proof. Call a window state *admissible* for coordinate j if it equals $\mathcal{W}_j(u)$ for some $u \in I_\varepsilon$; by Lemmas A.9 and A.10 there are at most S_n admissible states per coordinate and each has $k \geq K_n \geq 2$, so L_k and $V_{(k+1)}$ are well defined on every one of them. (States realized only outside I_ε may have smaller windows; they are not needed.) Define the exceptional events $T_1 = D_n(\eta)^c$; $T_2 = \{\exists i, j : V_{ij} \notin [\tau, 1 - \tau]\}$; $T_3 = \{\exists j, \text{ admissible state } \mathcal{W} : |L_k - 1| > t\}$; $T_4 = \{\exists j, \text{ admissible state } \mathcal{W} : V_{(k+1)} > 3\alpha\}$.

Outside $T_1 \cup T_2 \cup T_3 \cup T_4$, fix j and $u \in I_\varepsilon$: the realized window is an admissible state, and Lemma A.13 with $\xi_j \leq \Gamma$, $|L_k - 1| \leq t$, $L_k \leq 1 + t$ gives the displayed deviation bound; the modulus $\omega_n(h + \eta + \delta_n, \tau)$ covers every selected observation because its definition allows the displaced point $v = U_{ij}$ to range over all of $[0, 1]$.

Probabilities: $\mathbb{P}(T_1) \leq 2pe^{-2n\eta^2}$ is Lemma A.8; $\mathbb{P}(T_2) \leq 2pn\tau$ since each V_{ij} is uniform. For T_3, T_4 , condition on $\mathbf{U}_j$; the admissible states are then deterministic with $k \geq K_n$, Lemma A.12

applies to each, and Lemmas A.2 and A.3 give conditional bounds $2e^{-kt^2/4} \leq 2e^{-K_nt^2/4}$ and $e^{-(k+1)/4} \leq e^{-(K_n+1)/4}$, both decreasing in k . A union bound over the at most pS_n admissible states and integration over $\mathcal{U}_j$ complete the proof; overlapping states and dependent coordinates cost nothing beyond the union bound. $\square$

A.5 From profiles to scores

Recall the score grid $G = \{r\delta_n : r = 1, \dots, n\} \cap I_\varepsilon$, of size $L_n = |G|$, and that $K_n \geq 2$ makes every grid point usable, so $\hat{\Psi}_j = L_n^{-1} \sum_{u \in G} \hat{\xi}_j(u)$.

Lemma A.15 (Quadrature). *If $L_n \geq 1$, then for every j ,*

$$\left| L_n^{-1} \sum_{u \in G} \xi_j(u) - \Psi_j \right| \leq q_n := \omega_\xi(\delta_n) + \frac{4\Gamma\delta_n}{1-2\varepsilon}. \quad (\text{A.5})$$

Proof. Write the grid points $u_1 < \dots < u_{L_n}$, consecutive multiples of δ_n . If $L_n = 1$, then I_ε contains a single multiple of δ_n , which forces $|I_\varepsilon| < 2\delta_n$; the left side is at most Γ , while $4\Gamma\delta_n/|I_\varepsilon| > 2\Gamma$, so (A.5) holds trivially. Assume $L_n \geq 2$ and partition I_ε exactly: $C_1 = [\varepsilon, u_1 + \delta_n/2]$, $C_{L_n} = [u_{L_n} - \delta_n/2, 1 - \varepsilon]$, and $C_r = [u_r - \delta_n/2, u_r + \delta_n/2]$ otherwise. Interior cells have length δ_n , boundary cells length in $[\delta_n/2, 3\delta_n/2]$, and every point of C_r is within δ_n of u_r . Then

$$\left| \frac{1}{L_n} \sum_r \xi_j(u_r) - \frac{1}{|I_\varepsilon|} \int_{I_\varepsilon} \xi_j \right| \leq \frac{1}{|I_\varepsilon|} \sum_r \int_{C_r} |\xi_j - \xi_j(u_r)| + \Gamma \sum_r \left| \frac{\lambda(C_r)}{|I_\varepsilon|} - \frac{1}{L_n} \right|.$$

The first term is at most $\omega_\xi(\delta_n)$. For the second, both weight vectors sum to one; $|I_\varepsilon| = \sum_r \lambda(C_r) \in [(L_n - 1)\delta_n, (L_n + 1)\delta_n]$ gives $||I_\varepsilon|/L_n - \delta_n| \leq \delta_n/L_n$, so interior cells contribute at most δ_n in total and the two boundary cells at most $\delta_n/2 + \delta_n/L_n \leq \delta_n$ each; the sum is at most $3\delta_n$, and the second term is at most $3\Gamma\delta_n/|I_\varepsilon| \leq 4\Gamma\delta_n/|I_\varepsilon|$. $\square$

Proposition A.16 (Finite-sample score bound). *Under the conditions of Proposition A.14 (with in addition $\omega_\xi(\delta_n) < \infty$) and $L_n \geq 1$, for every $\eta > 0$, $0 < t \leq 1$, $0 < \tau < 1/2$,*

$$\mathbb{P}\left\{ \max_{j \leq p} |\hat{\Psi}_j - \Psi_j| > B_n(\eta, t, \tau) + q_n \right\} \leq 2pe^{-2n\eta^2} + 2pn\tau + 2pS_ne^{-K_nt^2/4} + pS_ne^{-(K_n+1)/4}, \quad (\text{A.6})$$

where $B_n(\eta, t, \tau) = 2\omega_n(h + \eta + \delta_n, \tau) + \Gamma t + (1+t)b_n(3\alpha)$.

Proof. Pathwise,

$$|\hat{\Psi}_j - \Psi_j| \leq \sup_{u \in I_\varepsilon} |\hat{\xi}_j(u) - \xi_j(u)| + \left| L_n^{-1} \sum_{u \in G} \xi_j(u) - \Psi_j \right|,$$

and Lemma A.15 bounds the second term by q_n on every sample path; reusing the same observations for estimation and averaging therefore costs no extra probability. Proposition A.14 gives (A.6). $\square$

Lemma A.17 (Deterministic score separation). *Suppose $\Delta_{\min} > 0$ and, on some event,*

$$\max_{j \leq p} |\hat{\Psi}_j - \Psi_j| < \frac{\Delta_{\min}}{2}.$$

Then, on that event, every active coordinate has a strictly smaller score than every inactive coordinate; consequently $A \subseteq \hat{A}_d$ for every $d \geq s$, and $\hat{A}_s = A$.

Proof. This lemma uses only the fibre identity (3), proved in Appendix B without reference to this subsection, through its consequence that inactive coordinates have $\Psi_j = \gamma^*$ (Proposition 2.2, claim 1); no circularity arises. Hence $\hat{\Psi}_j > \gamma^* - \Delta_{\min}/2$; active coordinates have $\Psi_j \leq \gamma^* - \Delta_{\min}$, hence $\hat{\Psi}_j < \gamma^* - \Delta_{\min}/2$. Every active score therefore precedes every inactive score in the increasing order, and the screening and recovery statements follow from the definition of $\hat{A}_d$. $\square$

B Proofs of the main results

This appendix proves the propositions of Section 2 and the theorems of Section 3 from the auxiliary results of Appendix A.

B.1 Proof of Proposition 2.1

For fixed $t > 0$, with M and $K_y^\star$ as in Lemmas A.4 and A.5 applied to $\mathcal{G}$,

$$\frac{M(ty)}{M(y)} = \int_{[\underline{\gamma}, \Gamma]} t^{-1/g} K_y^\star(dg) \longrightarrow t^{-1/\xi}$$

almost surely, by Lemma A.5 and boundedness and continuity of $g \mapsto t^{-1/g}$ on $[\underline{\gamma}, \Gamma]$. Lemma A.4 transfers the limit to $\mathbb{P}(Y > ty \mid \mathcal{G})/\mathbb{P}(Y > y \mid \mathcal{G})$. Convergence simultaneously for all $t > 0$ follows from rational t , monotonicity of the ratio in t , and continuity of $t \mapsto t^{-1/\xi}$.

For the fibre identity (3), fix $u \in I_\varepsilon$ and apply the above on the probability space $([0, 1]^{p-1}, K_j(u, \cdot))$ with the trivial σ -field; $\xi_j(u)$ is by definition the essential supremum of $v \mapsto \gamma(u, v)$ under $K_j(u, \cdot)$. Let $m = \max_v \gamma(u, v)$, attained by compactness and continuity. The essential supremum is at most m . Conversely, for $\eta > 0$ the set $\{v : \gamma(u, v) > m - \eta\}$ is open and nonempty, so has positive $K_j(u, \cdot)$ -probability under (S); the essential supremum is at least $m - \eta$ for every $\eta > 0$. $\square$

B.2 Proof of Proposition 2.2

Claim 1. For $j \notin A$, $\gamma(u, v) = g\{(u, v)_A\}$ does not involve u , and as v ranges over the fibre the active subvector ranges over all of $[0, 1]^s$; by (3), $\xi_j(u) = \max_w g(w) = \gamma^\star$ for every u .

Claim 2. By (3), $\xi_j(u) = \gamma^\star$ exactly when the fibre through u intersects M , i.e. when $u \in \pi_j(M)$.

Claim 3. $u \mapsto \gamma^\star - \xi_j(u)$ is continuous (Lemma A.6), nonnegative, and by Claim 2 vanishes on I_ε exactly on $I_\varepsilon \cap \pi_j(M)$. A continuous nonnegative function has positive integral over I_ε if and only if its positivity set has positive Lebesgue measure.

Final statement. Write M^A for the maximizer set of g in $[0, 1]^s$; then $M = \{u : u_A \in M^A\}$ and, for $j \in A$, $\pi_j(M)$ is the projection of M^A , a finite set when M^A is finite; its intersection with I_ε is Lebesgue-null and Claim 3 applies. $\square$

B.3 Proof of Proposition 2.3

Every x on the fibre $x_j = u$ has $\|x_A - u_A^\star\| \geq |u - u_j^\star|$, so every point of the fibre satisfies $\gamma(x) \leq \gamma^\star - c|u - u_j^\star|^\nu$; taking the fibre maximum gives $\xi_j(u) \leq \gamma^\star - c|u - u_j^\star|^\nu$. Averaging over $u \in I_\varepsilon$ gives the score bound. For $\nu = 2$, $\mathbb{E}(V - u_j^\star)^2 = \text{Var}(V) + (1/2 - u_j^\star)^2 = (1 - 2\varepsilon)^2/12 + (1/2 - u_j^\star)^2$. $\square$

B.4 Proof of Theorem 3.1

First, the standing hypotheses of Propositions A.14 and A.16 hold for n large: $\log(pn) = o(nh^2)$ with $h \leq 1$ forces $nh \rightarrow \infty$, hence $m_n^- \geq nh/2 \rightarrow \infty$ and $L_n \geq 1$; $\log(pn) = o(nah)$ then forces $\alpha m_n^- \geq nah/2 \rightarrow \infty$, hence $K_n \geq 2$; and $\omega_\xi(\delta_n) < \infty$ by (E1). Choose $\eta_n = \sqrt{\log(pn)/n}$, $\tau = \tau_n = (pn)^{-2}$, and $t_n = \sqrt{(4/K_n) \log\{pS_n(pn)\}}$. The four probability terms in (A.6) are then each $o(1)$: the first is $2pe^{-2\log(pn)} = 2p(pn)^{-2} \rightarrow 0$; the second is $2pn(pn)^{-2} \rightarrow 0$; the third is $2pS_n\{pS_n(pn)\}^{-1} = 2(pn)^{-1} \rightarrow 0$; the fourth is $pS_n e^{-(K_n+1)/4} \rightarrow 0$ because $\log(pS_n) \leq C \log(pn) = o(K_n)$ by the second condition in (7) and Lemma A.11. The same condition makes $t_n \leq 1$ for large n , as required by Lemma A.2, and $\Gamma t_n = O(\sqrt{\log(pn)/(nah)})$, again by Lemma A.11.

It remains to check that the bias part of $B_n(\eta_n, t_n, \tau_n) + q_n$ in Proposition A.16 is $o(\Delta_{\min})$. First, $\log(pn) = o(nh^2)$ gives $\eta_n + \delta_n = o(h)$, so $h + \eta_n + \delta_n \leq C_0 h$ eventually and $2\omega_n(h + \eta_n + \delta_n, \tau_n) \leq 2\omega_n(C_0 h, \tau_n) = o(\Delta_{\min})$ by (E3) and monotonicity of $\omega_n(\cdot, \tau)$. Second, $(1 + t_n)b_n(3\alpha) \leq 2b_n(3\alpha) = o(\Delta_{\min})$ by (E2). Third, $q_n = \omega_\xi(\delta_n) + O(1/n)$: the first part is $o(\Delta_{\min})$ by (E1), and $1/n \leq \sqrt{\log(pn)/(n\alpha h)}$ since $\alpha h \leq 1 \leq n \log(pn)$, so the second part is absorbed into the stochastic rate. Collecting terms, outside a vanishing probability,

$$\max_{j \leq p} |\hat{\Psi}_j - \Psi_j| \leq a_n + b'_n, \quad a_n = O\left(\sqrt{\log(pn)/(n\alpha h)}\right), \quad b'_n = o(\Delta_{\min}),$$

which is the assertion. $\square$

B.5 Proof of Theorem 3.2 and Corollary 3.3

Under (8), $\sqrt{\log(pn)/(n\alpha h)} = o(\Delta_{\min})$, so the bound of Theorem 3.1 is below $\Delta_{\min}/2$ for n large, outside a vanishing probability. On the complementary event, Lemma A.17 places every active score strictly before every inactive score, so $A \subseteq \hat{A}_d$ for every $d \geq s$, and $\hat{A}_s = A$ when $d = s$. $\square$

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